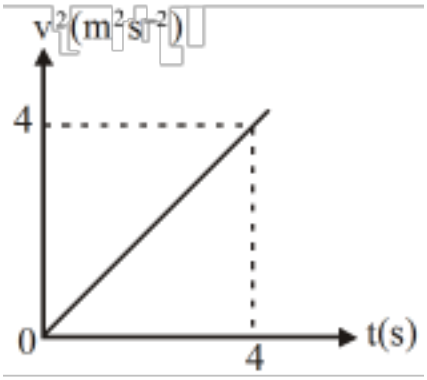
PHYSICS

1) A particle is moving along a straight line such that square of its velocity varies with time as shown in the figure. What is the acceleration of the particle at t = 4 s ?



(1) 4 m/s^2

(2) 1/4 m/s^2

(3) 1/2 m/s^2

(4) 0

2) In a straight line motion particle start from rest and travel with constant acceleration covers distance x in first 3 seconds and y in next 9 seconds then ratio $\frac{y}{x}$ equals?

(1) 1 : 15

(2) 1 : 9

(3) 9 : 1

(4) 15 : 1

3) The external and internal diameters of a hollow cylinder are measured to be (4.23 ± 0.01) cm and (3.89 ± 0.01) cm. The thickness of the wall of the cylinder is :-

(1) (0.34 ± 0.02) cm

(2) (0.17 ± 0.02) cm

(3) (0.17 ± 0.01) cm

(4) (0.34 ± 0.01) cm

4) Match the physical quantities mentioned in **Column I** with their respective dimensional given in **Column II.**

	Column I		**Column II**
A	Inductance (L)	P	$[M^1L^2T^{-2}K^{-1}]$
B	Boltzmann's constant (k)	Q	$[M^1L^3T^{-3}A^{-2}]$

C	Magnetic permeability (μ_0)	R	$[M^1L^2T^{-2}A^{-2}]$
D	Resistivity (ρ)	S	$[M^1L^1T^{-2}A^{-2}]$

(1) A-Q, B-S, C-P, D-R

(2) A-S, B-P, C-Q, D-R

(3) A-R, B-P, C-S, D-Q

(4) A-Q, B-P, C-R, D-Q

5) A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pully at the edge of table and from its other end another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and the table is μ_k. When the block A is sliding on the table, the tension in the string is

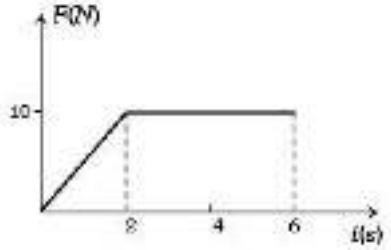
(1) $\frac{(m_2 + \mu_k m_1)g}{m_1 + m_2}$

(2) $\frac{(m_2 - \mu_k m_1)g}{m_1 + m_2}$

(3) $\frac{m_1 m_2 (1 + \mu_k)g}{m_1 + m_2}$

(4) $\frac{m_1 m_2 (1 - \mu_k)g}{m_1 + m_2}$

6) A body of mass 3kg is acted on by a force which varies as shown in the graph below. The momentum acquired is given by ..... N – s.



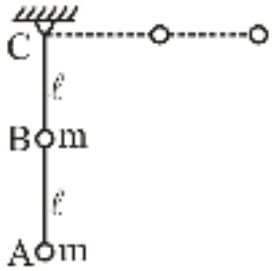
(1) 0

(2) 5

(3) 30

(4) 50

7) A weightless rod of length 2ℓ carries two equal masses 'm', one tied at lower end A and the other at the middle of the rod at B. The rod can rotate in vertical plane about a fixed horizontal axis passing through C. The rod is released from rest in horizontal position. The speed of the mass B at the instant rod, become vertical is :-



(1) $\sqrt{\frac{3g\ell}{5}}$

(2) $\sqrt{\frac{4g\ell}{5}}$

(3) $\sqrt{\frac{6g\ell}{5}}$

(4) $\sqrt{\frac{7g\ell}{5}}$

8) The gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Considering only translational and rotational modes, the total internal energy of system is-

(1) 12 RT

(2) 20 RT

(3) 15 RT

(4) 4 RT

9) Two tanks A and B contain water at 30°C and 80°C respectively. The amount of water that must be taken from A and B to prepare 40 kg water at 50°C.

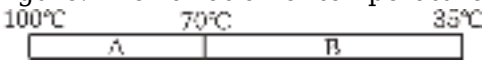
(1) 24 kg, 16 kg

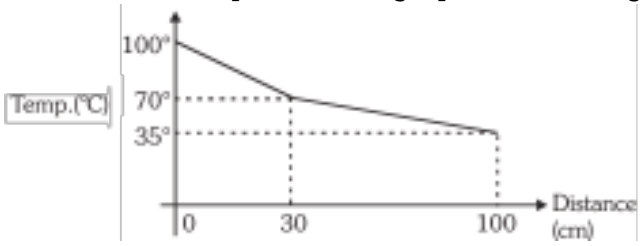
(2) 16 kg, 24 kg

(3) 20 kg, 20 kg

(4) 30 kg, 10 kg

10) Two different rods A and B are kept as shown in figure. The variation of temperature of different cross sections is plotted in a graph shown in figure.





The ratio of thermal conductivities of A and B is :

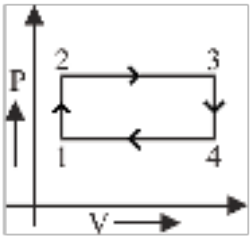
(1) 2

(2) 0.5

(3) 1

(4) 2/3

11) Three moles of an ideal monoatomic gas perform a cycle as shown in the figure. The gas temperature in different states are : T_1 = 400K, T_2 = 800 K, T_3 = 2400 K and T_4 = 1200 K. The of

work done by the gas during the cycle is: 

(1) 10 kJ

(2) 20 kJ

(3) 5 kJ

(4) 8.3 kJ

12) In a resonance tube experiment, the first resonance is obtained for 10 cm of air column and the second resonance at 32 cm. The end correction for this apparatus is–

(1) 0.5 cm

(2) 1.0 cm

(3) 1.5 cm

(4) 2 cm

13) A body of mass 36 gm moves with SHM of amplitude A = 13 cm and time period T = 12 sec. At time t = 0, the displacement x is +13 cm. The shortest time of passage from x = +6.5 cm to x = –6.5 cm is :-

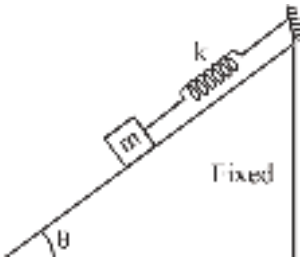
(1) 4 sec

(2) 2 sec

(3) 6 sec

(4) 3 sec

14) In the figure shown, the time period and the amplitude respectively, when block of mass m is released from rest when the spring is relaxed is :- (the inclined plane is smooth)



(1) $2\pi\sqrt{\frac{m}{k}}, \frac{mg\sin\theta}{k}$

(2) $2\pi\sqrt{\frac{m\sin\theta}{k}}, \frac{2mg\sin\theta}{k}$

(3) $2\pi\sqrt{\frac{m}{k}}, \frac{mg\cos\theta}{k}$

(4) None of these

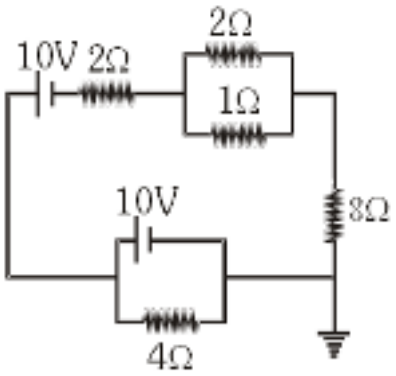
15) Electric potential at any point is $V = -5x + 3y + \sqrt{15}z$, then the magnitude of the electric field is

: (in proper unit)

(1) $3\sqrt{2}$
(2) $4\sqrt{2}$
(3) $5\sqrt{2}$
(4) 7

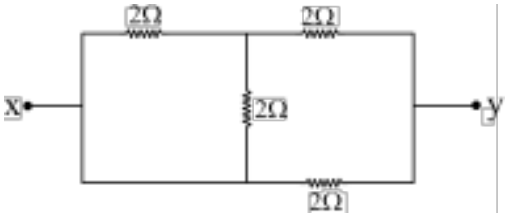
16) Non conducting ring of radius R has uniformly distributed positive charge Q. A small part of the ring, of length d, is removed (d << R). The electric field at the centre of the ring will now be :-

(1) directed towards the gap, inversely proportional to R^3
(2) directed towards the gap, inversely proportional to R^2
(3) directed away from the gap, inversely proportional to R^3
(4) directed away from the gap, inversely proportional to R^2



17) In the circuit shown, the current in 4Ω, 8Ω resistances are -

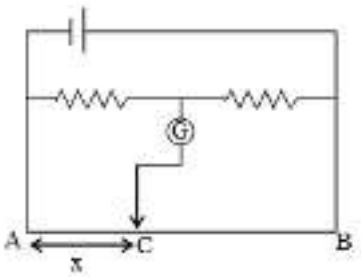
(1) 2.5 A, 0A
(2) 5A, 0A
(3) 2.5A, $\frac{40}{3}$ A
(4) 5A, $\frac{80}{3}$ A



18) Effective resistance between x and y will be ?

(1) 5 ohm
(2) $\frac{6}{5}$ ohm
(3) $\frac{5}{6}$ ohm
(4) 10 ohm

19) In this given circuit, no current is passing through the galvanometer. If the cross-sectional diameter of the wire AB is doubled then for null point of galvanometer the value of AC would–



(1) $\frac{x}{4}$

(2) 4x

(3) 2x

(4) x

20) Two capacitors whose capacitances and break down voltages are given as follows(C_1 = 2μF, V_1 = 6KV), (C_2 = 4μF, V_2 = 2KV) If they are connected in series then the combination can with stand a voltage of:

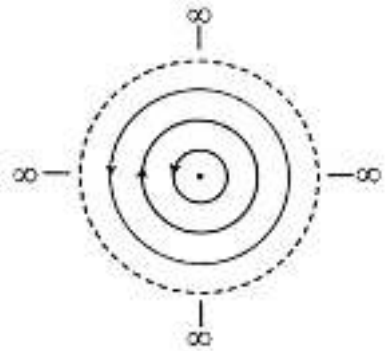
(1) 10 KV

(2) 9 KV

(3) 8 KV

(4) 5 KV

21) In figure, infinite conducting rings each having current i in the direction shown are placed concentrically in the same plane as shown in the figure. The radius of rings are r, 2r, 2^2r,



2^3r.....∞. The magnetic field at the centre of rings will be

(1) Zero

(2) $\frac{\mu_0 i}{r}$

(3) $\frac{\mu_0 i}{2r}$

(4) $\frac{\mu_0 i}{3r}$

22) The area of hysteresis loop of a material is equivalent to 250 m^2. When 10 kg material is magnetised by an alternating field of 50 Hz then energy lost in one hour will be if the density of material is 7.5 gm/cm^3 :

(1) 6×10^4 J

(2) 6×10^4 erg

(3) 3×10^2 J

(4) 3×10^2 erg

23) If a variable frequency ac source is connected to a capacitor then with decrease in frequency, the displacement current will

(1) increase

(2) decrease

(3) remains constant

(4) first decrease then increase

24) If 4 cm and 2 cm be the sizes of the images respectively for the two positions of the lens in the displacement method, then the size of the object given by :

(1) $\sqrt{2}$ cm

(2) $2\sqrt{2}$ cm

(3) 2 cm

(4) 8 cm

25) The angle of deviation produced by a thin glass prism when placed in air is 'δ_1' and that when immersed in water is 'δ_2'. The refractive index of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. The ratio $\delta_1 : \delta_2$ is -

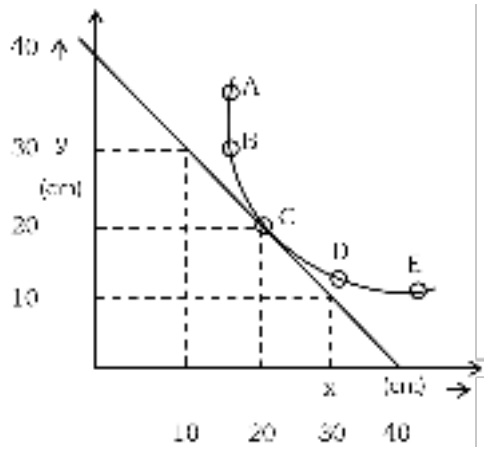
(1) 1 : 2

(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

26) In an experiment to measure the focal length (f) of a convex lens, the magnitude of object distance (x) and the image distance (y) are measured with reference to the focal point of the lens. The y-x plot is shown in figure.



The focal length of the lens is _____ cm.

(1) 10

(2) 20

(3) 30

(4) 40

27) In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the



plane of the slits so this is closer to the upper slit, as shown. Then -

(1) The fringe width will increase and fringe pattern will shift down.

(2) The fringe width will remain same but fringe pattern will shift up.

(3) The fringe width will decrease and fringe pattern will shift down.

(4) The fringe width will remain same but fringe pattern will shift down.

28) Two polaroids P_1 and P_2 are placed with their axis parallel to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 30° with that of P_1. The intensity of transmitted light through P_2 is :-

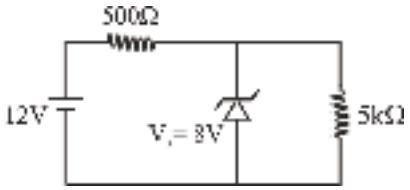
(1) $\frac{I_0}{8}$

(2) $\frac{3I_0}{32}$

(3) $\frac{9I_0}{32}$

(4) $\frac{I_0}{32}$

29) Power lost in zener diode in the given circuit, is:-



(1) 12.8 mW

(2) 32 mW

(3) 51.2 mW

(4) 96 mW

30) When forward bias is applied to a P-N Junction then what happens to the potential barrier V_B and the width of depletion region X :-

(1) V_B increases, X decrease.

(2) V_B decreases, X increases.

(3) V_B increases, X increases

(4) V_B decreases, X decreases

31) A particle performs S.H.M. of amplitude A with angular frequency ω along a straight line. When it is at a distance $\frac{\sqrt{3}}{2}$A from mean position, its kinetic energy gets increased by an amount $\frac{1}{2}m\omega^2A^2$ due to an impulsive force. Then its new amplitude becomes-

(1) $\frac{\sqrt{5}}{2}A$

(2) $\frac{\sqrt{3}}{2}A$

(3) $\sqrt{2}A$

(4) $\sqrt{5}A$

32) The equation of a wave is represented by $y = 10^{-4}\sin\left[100t - \frac{x}{10}\right]$. where t is in second and y, x is in meter. The velocity of the wave will be :-

(1) 100 m/s

(2) 250 m/s

(3) 750 m/s

(4) 1000 m/s

33) Regarding speed of sound in gas match the following :-

(A)	Temperature of gas is made 4 times and pressure 2 times	(P)	Speed becomes $2\sqrt{2}$times
(B)	Only pressure is made 4 times without change in temperature	(Q)	Speed become 2 times
(C)	Only temperature is changed to 4 times	(R)	Speed remains unchanged
(D)	Molecular mass of the gas is made 4 times	(S)	Speed become half

(1) A – Q, B – R, C – Q, D – S

(2) A – Q, B – S, C – Q, D – S

(3) A – S, B –R, C – Q, D – S

(4) A – Q, B – R, C – Q, D – Q

34) A wire of initial length L and radius r is stretched by a length □. Another wire of same material but with initial length 2L and radius 2r is stretched by a length 2□. The ratio of the stored elastic energy per unit volume in the first and second wire is :

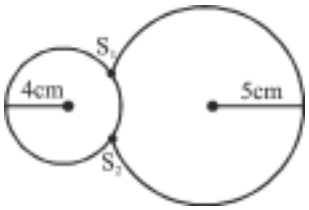
(1) 1 : 4

(2) 1 : 2

(3) 2 : 1

(4) 1 : 1

35) Two soap bubbles of radii r_1 and r_2 equal to 4cm and 5cm are touching each other over a common surface S_1S_2 (shown in figure). What is the radius of common surface?



(1) 4 cm

(2) 20 cm

(3) 5 cm

(4) 4.5cm

36) Maximum height reached by an object projected perpendicular to the surface of the earth with a speed equal to 50% of the escape velocity from earth surface is - (R = Radius of Earth) :-

(1) $\frac{R}{2}$

(2) $\frac{16R}{9}$

(3) $\frac{R}{3}$

(4) $\frac{R}{8}$

37)

Find the distance from center of a uniform solid sphere at which the gravitational potential is half of the potential at center of sphere.
(R = Radius of sphere)

(1) $\sqrt{\frac{3}{2}}R$

(2) $\frac{R}{3}$

(3) $\frac{4R}{5}$

(4) $\frac{4R}{3}$

38) The moment of inertia of a solid cylinder about its own axis is the same as its moment of inertia about an axis passing through its centre of gravity and perpendicular to its length. The relation between its length L and radius R is-

(1) $L = \sqrt{2}R$

(2) $L = \sqrt{3}R$

(3) L = 3R

(4) L = R

39) A satellite of mass m revolving around the planet of mass M in a circular orbit with orbital velocity V_0, then the relation between its time period (T) and orbital velocity is :-

(1) $T = \frac{2\pi GM}{V_0^2}$

(2) $T = \frac{\pi GM}{V_0^3}$

(3) $T = \frac{GM}{V_0^3}$

(4) $T = \frac{2\pi GM}{V_0^3}$

40) If $\vec{H} = -2\cos(\omega t - kz)\hat{i}$ in an EMW propagating along +z axis then electric field is given by

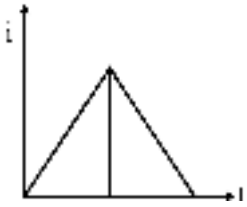
(1) $\vec{E} = 753\cos(\omega t - kz)\hat{j}$

(2) $\vec{E} = 387\sin(\omega t - kz)\hat{j}$

(3) $\vec{E} = 753\cos(\omega t + kz)\hat{j}$

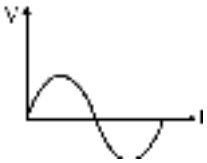
(4) $\vec{E} = -753\sin(\omega t + kz)\hat{j}$

41) The current 'i' in an inductance coil varies with time 't' according to following graph

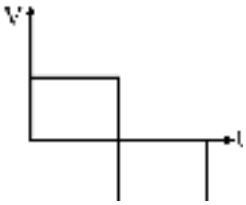


Which one of the following plots shows the variations of voltage in the coil

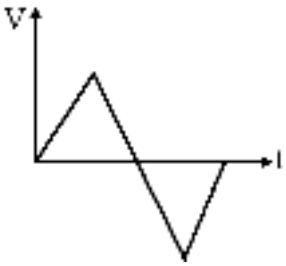
(1)

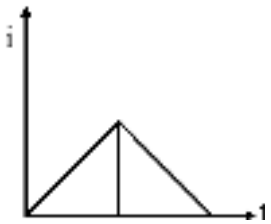


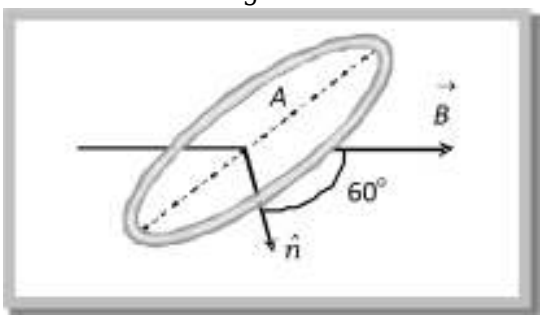
(2)



(3)



(4) 

42) A coil of area A = 0.5 m^2 is situated in a uniform magnetic field B = 4.0 wb/m^2 and area vector makes an angle of 60° with respect to the magnetic field as shown in figure. The value of the magnetic flux through the area A would be equal to 

(1) 2 weber

(2) 1 weber

(3) 3 weber

(4) $\frac{3}{2}$ weber

43) A mass 'm' moves with a velocity 'v' and collides inelastically with another identical mass in rest. After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision.

(1) v

(2) $\sqrt{3}\,v$

(3) $\frac{2}{\sqrt{3}}v$

(4) $\frac{v}{\sqrt{3}}$

44) A stepdown transformer reduce the voltage of transmission line from 2200 V to 220 V. The power delivered by it is 880 W and its efficiency is 88%. The input current is :-

(1) 4.65 mA

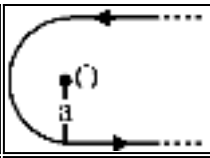
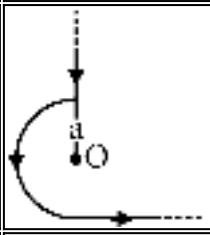
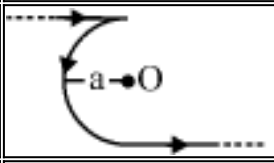
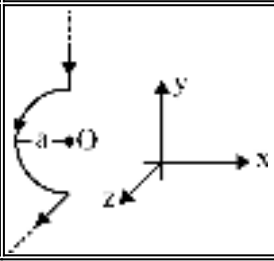
(2) 0.465 mA

(3) 0.4545 A

(4) 4.65 A

45)

List-I	List-II
Current Carrying Wires	**B_0**

(P)		(1)	$\frac{\mu_0 i}{4a}\left(1+\frac{1}{\pi}\right)$
(Q)		(2)	$\frac{\mu_0 i}{4a}\sqrt{1+\frac{1}{\pi^2}}$
(R)		(3)	$\frac{\mu_0 i}{2a}\left(\frac{1}{2}+\frac{1}{\pi}\right)$
(S)		(4)	$\frac{\mu_0 i}{4a}$

(1) P-3, Q-1, R-4, S-2

(2) P-4, Q-2, R-3, S-1

(3) P-2, Q-3, R-1, S-4

(4) P-1, Q-4, R-2, S-3

CHEMISTRY

1)

Among the following species identify the isostructural pairs:-
NF_3, NO_3^-,BF_3, H_3O^+, HN_3

(1) [NF_3, NO_3^-] and [BF_3, H_3O^+]

(2) [NF_3, HN_3] and [H_3O^+,BF_3]

(3) [NF_3, H_3O^+] and [NO_3^- , BF_3]

(4) [NF_3, H_3O^+] and [HN_3, BF_3]

2) In O_2^-, O_2 and O_2^{2-}, the total number of antibonding electrons respectively are

(1) 7, 6, 8

(2) 1, 0, 2

(3) 6, 6, 6

(4) 8, 6, 8

3) The strongest H–Bond ?

(1) 

(2) 

(3) 

(4) 

4) Which of the following is not correct order for dipole moment?

(1) HF > HCl > HBr > HI

(2) $H_2O > H_2S > CO_2$

(3) $CHCl_3 > CH_2Cl_2 > CH_3Cl$

(4) $CHCl_3 > H_2S > HBr > NF_3$

5) Identify the **incorrect** match :

	Name		**IUPAC Official Name**
(a)	Unnilunium	(i)	Mendelevium
(b)	Unniltrium	(ii)	Lawrencium
(c)	Unnilhexium	(iii)	Seaborgium
(d)	Unununnium	(iv)	Darmstadtium

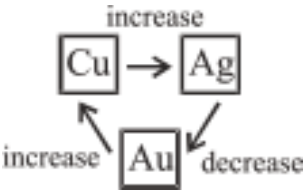
(1) (d), (iv)

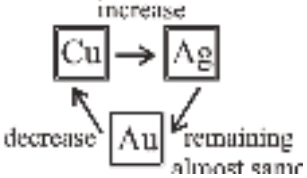
(2) (a), (i)

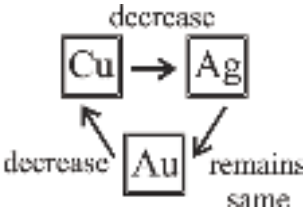
(3) (b), (ii)

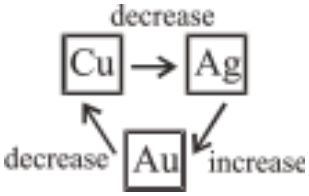
(4) (c), (iii)

6) Which diagram of ionization energy is correct ?

(1) 

(2) 

(3) 

(4) 

7)

The IUPAC name of Xe[Pt F_6] is :

(1) Hexafluoroplatinate (VI)xenon
(2) Xenonhexafluoroplatinate (V)
(3) Xenonhexafluoroplatinate (VI)
(4) Xenoniumhexafluoroplatinum(V)

8) If $P > \Delta_0$ then electronic configuration of d^5 in octahedral complex :-

(1) $t_{2g}^5 eg^0$
(2) $t_{2g}^3 eg^2$
(3) $t_{2g}^4 eg^1$
(4) $eg^5 t_{2g}$

9)

	Compound		**Total stereoisomers**
(A)	$[Co(en)_2Br_2]^+$	(M)	2
(B)	$K_3[Co(OX)_3]$	(N)	3
(C)	$[Pt(NH_3)_2Cl_2]$	(O)	4
(D)	$[Pt(NH_3)_3Br_3]^+$	(P)	5

Correct is :

	(A)	(B)	(C)	(D)
(1)	O	M	O	O
(2)	N	M	M	M
(3)	M	O	M	M
(4)	N	N	M	M

(1) 1
(2) 2
(3) 3
(4) 4

10) ***Assertion (A)*** :- Fe^{+3} is not used in brown ring test of NO_3^{-1}. ***Reason (R)*** :- Reduction of NO_3^-

takes place during test.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

11) Match the catalyst with the process :-

Catalyst		**Process**	
(i)	V_2O_5	(a)	The oxidation of ethyne to ethanal
(ii)	$TiCl_4 + Al(CH_3)_3$	(b)	Polymerisation of alkynes
(iii)	$PdCl_2$	(c)	Oxidation of SO_2 in the manufacture of H_2SO_4
(iv)	Nickel complexes	(d)	Polymerisation of ethylene

Which of the following is the correct option ?

(1) i-c, ii-d, iii-a, iv-b

(2) i-a, ii-b, iii-c, iv-d

(3) i-a, ii-c, iii-b, iv-d

(4) i-c, ii-a, iii-d, iv-b

12) Which of the following does not give chromyl chloride test :-

(1) NaCl

(2) KCl

(3) $HgCl_2$

(4) NH_4Cl

13) **Statement (I) :** Boron form only covalent compounds
Statement (II) : The sum of its first three ionisation enthalpies is very high

(1) Both statements I and II are correct, statement II is the correct explanation of statement I.

(2) Both statements I and II are correct but statement II is the not correct explanation of statement I.

(3) Statement I is correct and Statement II is incorrect

(4) Both statements I and II are incorrect

14)

Choose the correct matching :-

	Column-I (Given Mixture)		**Column-II (Colour Precipitate)**

(P)	Pb^{+2} + dil HCl + H_2S→	(A)	Rosy Red ppt.
(Q)	Fe^{+3} + NH_4OH →	(B)	White ppt.
(R)	Ni^{+2} + DMG solution →	(C)	Black ppt.
(S)	Zn^{+2} + NH_4OH →	(D)	Brown ppt.

(1) P-C, Q-D, R-A, S-B

(2) P-C, Q-D, R-B, S-A

(3) P-B, Q-C, R-A, S-D

(4) P-D, Q-C, R-A, S-B

15)

	List-I (Basic Radical)		**List-II (Group)**
(P)	Pb^{+2}	(1)	VI group
(Q)	Fe^{+3}	(2)	IV group
(R)	Ni^{+2}	(3)	I group
(S)	Mg^{+2}	(4)	III group

Correct matching is :

	P	**Q**	**R**	**S**
(1)	1	3	2	4
(2)	3	4	2	1
(3)	4	3	2	1
(4)	3	4	1	2

(1) 1

(2) 2

(3) 3

(4) 4

16) Match list-I with list-II :-

	List-I		**List-II**
(A)	2.8 g CO	I	0.1 N_A atoms
(B)	1.4 g N_2	II	N_A molecules
(C)	1 mol O_2	III	2.24 L at STP
(D)	0.5 mol CH_4	IV	3×10^{24} electrons

(1) A → II, B → IV, C → I, D → III

(2) A → II, B → III, C → I, D → IV

(3) A → III, B → II, C → I, D → IV

(4) A → III, B → I, C → II, D → IV

17) How many faradays are required to reduce 2 mol of MnO_4^- to Mn^{2+} ?

(1) 1

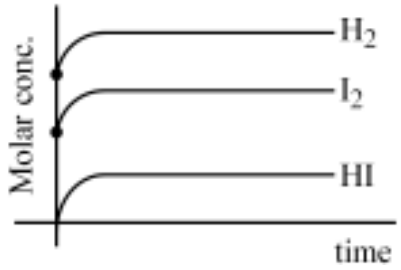
(2) 5

(3) 3

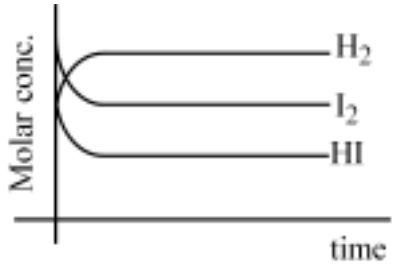
(4) 10

18) For the reaction $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$
Attainment of equilibrium is predicted correctly by :-

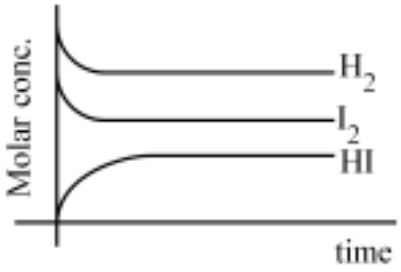
(1)



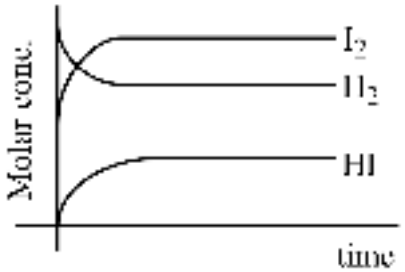
(2)



(3)



(4)



19) Λ_m° for NaCl, HCl and NaA are 126.4, 425.9 and 100.5 S cm^2 mol^{-1}, respectively. If the conductivity of 0.001 M HA is 5×10^{-5} S cm^{-1}, degree of dissociation of HA is

(1) 0.75

(2) 0.125

(3) 0.25

(4) 0.50

20) The plot of ln k vs $\frac{1}{T}$ for the following reaction
$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$
gives a straight line with the slope of line equal to -1.0×10^4 K. Activation energy for the reaction in J mol^{-1} is (Given R = 8.3 J K^{-1} mol^{-1})

(1) 4.0×10^{-2}

(2) 8.3×10^{-4}

(3) 8.3×10^{4}

(4) 4.0×10^{2}

21) K_{sp} of $Cr(OH)_3$ is 1.6×10^{-30}. What is the solubility of $Cr(OH)_3$ in water ?

(1) $\sqrt{1.6 \times 10^{-30}}$

(2) $\frac{1.6 \times 10^{-30}}{27}$

(3) $\left(\frac{1.6 \times 10^{-30}}{27}\right)^{1/4}$

(4) $\left(\frac{1.6 \times 10^{-30}}{4}\right)^{1/3}$

22) Van't Hoff factor of Hg_2Cl_2 in its aqueous solution will be if Hg_2Cl_2 is 80% ionized in the solution:

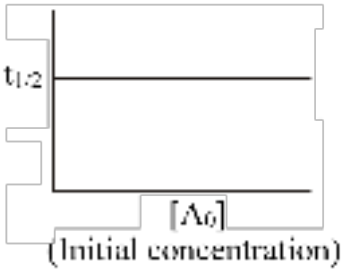
(1) 1.6

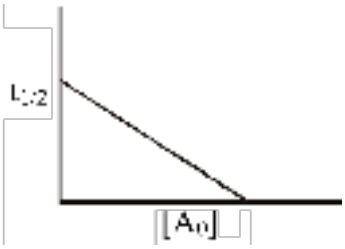
(2) 2.6

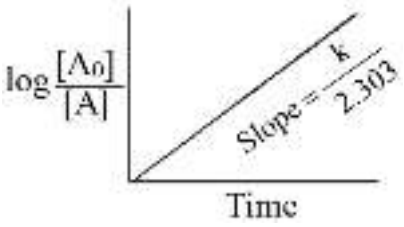
(3) 3.6

(4) 4.6

23) In the following graphs, which graph(s) belong to first order

(i) 

(ii) 

(iii) 

(1) (i), (iii)

(2) (i), (ii), (iii)

(3) (i) only

(4) (iii), (ii)

24) An aqueous solution of a solute AB has boiling point of 101.08°C (AB is 100% ionised at boiling point of the solution) and freezes at –1.80°C.
Hence, AB (K_b / K_f = 0.3)

(1) Is 100% ionised at the freezing point of the solution

(2) Behaves as non-electrolyte at the freezing point of the solution

(3) Forms dimer

(4) Is 50% ionised at freezing point of solution

25) Arrange the pH of following solution in decreasing order :-
(A) 0.01 M HCl
(B) 0.01 M NaOH
(C) 0.01 M CH_3COONa
(D) 0.01 M NaCl

(1) B > C > D > A

(2) A > D > C > B

(3) A > C > D > B

(4) B > D > C > A

26) **Assertion (A) :-** Work done in free expansion of ideal gas is zero.
Reason (R) :- In free expansion entropy remain constant.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct, Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

27) The hydration energy of K^+ & Cl^- are –x & –y $kJmol^{-1}$. The lattice enthalpy of KCl is z $kJmol^{-1}$ then heat of solution of KCl is :-

(1) x + y + z

(2) x + y – z

(3) z – x – y

(4) x – y – z

28) The standard reduction potentials of Zn, Ni, Cu and Ag are –0.76 V, –0.22 V, +0.34 V and +0.80 V respectively then which of the following cell reactions will be spontaneous :-

(I) $Ni_{(s)} + Zn^{+2}_{(aq)} \rightleftharpoons Ni^{+2}_{(aq)} + Zn_{(s)}$

(II) $Cu_{(s)} + Zn^{+2}_{(aq)} \rightleftharpoons Zn_{(s)} + Cu^{+2}_{(aq)}$

(III) $H_{2(g)} + 2Ag^{+}_{(aq)} \rightleftharpoons 2H^{+}_{(aq)} + 2Ag_{(s)}$

(IV) $Ni_{(s)} + 2Ag^{+}_{(aq)} \rightleftharpoons Ni^{+2}_{(aq)} + 2Ag_{(s)}$

(1) I, II

(2) I, III, IV

(3) I, IV

(4) III, IV

29)

The wavelength of photon 'A' is 400 nm. The frequency of photon 'B' is 10^{16} s^{-1}. The wave number of photon 'C' is 10^{4} cm^{-1}. The correct order of energy of these photons is :-

(1) A > C > B

(2) C > B > A

(3) B > A > C

(4) A > B > C

30) Number of moles of H^{+} ions involve in reaction when 1 mol of MnO_4^{-} to oxidise oxalate ion to CO_2 is :-

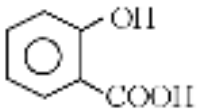
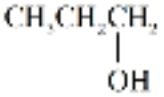
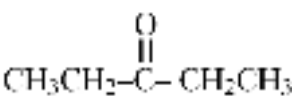
(1) 4

(2) 8

(3) 2

(4) 1

31) Match the following column :

Column-I (Compounds)		Column-II (distinction by)	
(A)	CH_3CHO and HCHO	(P)	
(B)	 and CH_3OH	(Q)	I_2 + NaOH
(C)	PhOH and CH_3COOH	(R)	$AgNO_3$ + NH_4OH
(D)	 and $CH_3(CH_2)_3CHO$	(S)	$NaHCO_3$

(1) A-Q,B-S,C-P, D-R

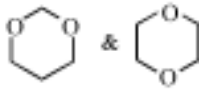
(2) A-Q, B-P, C-R, D-S

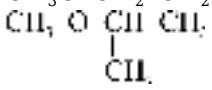
(3) A-Q,B-P,C-S, D-R

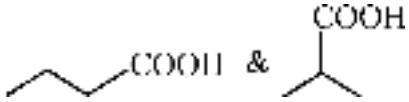
(4) A-R, B-P, C-S, D-Q

32) Which of following pairs of compounds are not position isomers ?

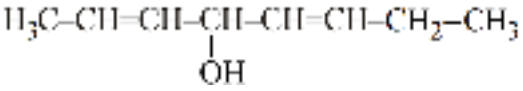
(1) 

(2) 

(3) $CH_3O–CH_2–CH_2–CH_3$ &


(4) 

33) How many stereoisomers are possible for the given compound

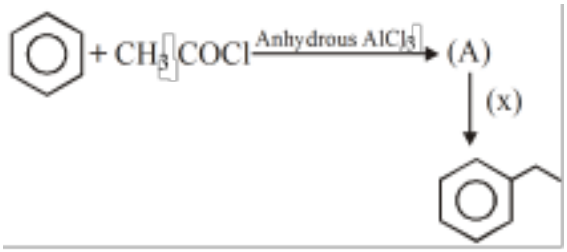


(1) 4

(2) 8

(3) 2

(4) 6

34) 

Reagent (x) can be :-

(1) HI / Red P

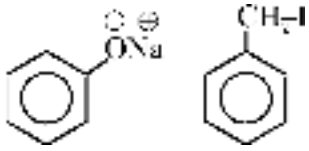
(2) Zn–Hg / HCl

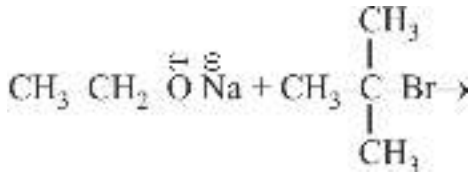
(3) $NH_2 – NH_2 / \overset{\ominus}{O}H/\Delta$

(4) All of the above

35) Which of the following will not give ether as product :-

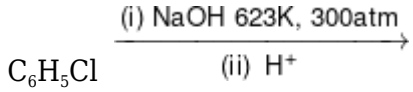
(1) $CH_3–CH_2–\overset{\ominus\oplus}{O}Na + CH_3–CH_2–Cl \rightarrow$

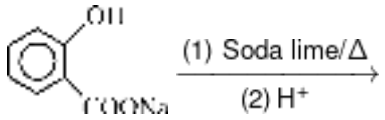
(2) 

(3) 

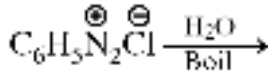
(4) All of these

36) Which of the following reaction does not give phenol?

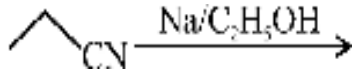
(1) 

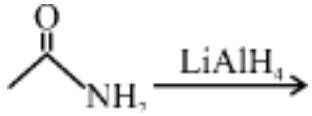
(2) 

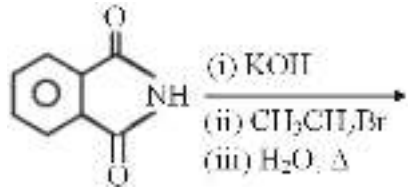
(3) 

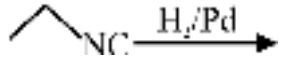
(4) 

37) Which of the following reaction does not give primary amine as a product ?

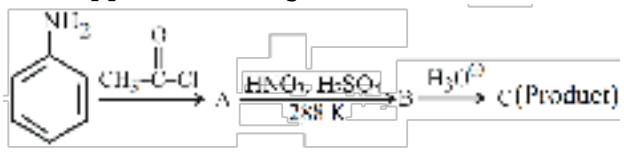
(1) 

(2) 

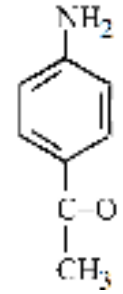
(3) 

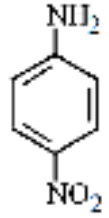
(4) 

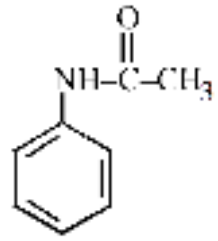
38) Suppose following reaction

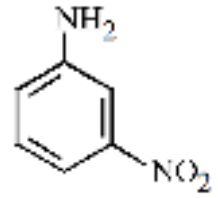


The major product 'C' will be :-

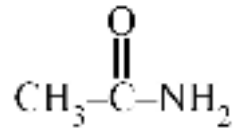
(1) 

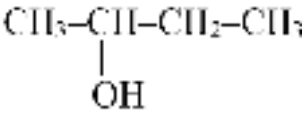
(2) 

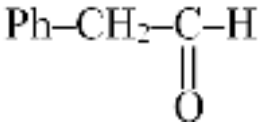
(3) 

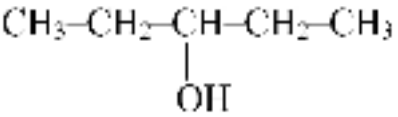
(4) 

39) Which of the following compound give Iodoform test ?

(1) 

(2) 

(3) 

(4) 

40) **Statement-I :-** During the esterification of carboxylic acid with alcohol, protonation of the carbonyl oxygen activates the carbonyl group towards N.A.R. of alcohol.
Statement-II :- Proton transfer in the tetrahedral intermediate convert the hydroxyl group into a better leaving group $-\overset{+}{O}H_2$.

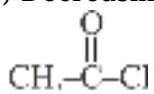
(1) Both statement-I and statement-II are correct.

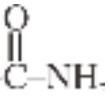
(2) Statement-I is correct but statement-II is incorrect.

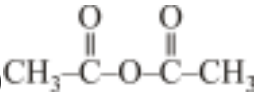
(3) Statement-I is incorrect but statement-II is correct.

(4) Both statement-I and statement-II is incorrect.

41) Decreasing order of reactivity of nucleophilic substitution reaction is :

(I) 

(II) 

(III) 

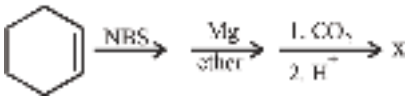
(IV) 

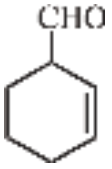
(1) I > II > III > IV

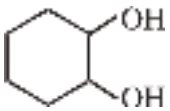
(2) IV > III > II > I

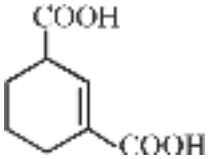
(3) II > III > I > IV

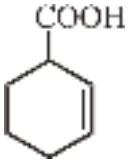
(4) I > III > IV > II

42) In the given reaction (x) will be :- 

(1) 

(2) 

(3) 

(4) 

43) Which will give same osazone when treated with phenyl hydrazine ?
(A) Glucose (B) Sucrose (C) Fructose
(D) Maltose (E) Mannose

(1) A, B, E

(2) A, C, E

(3) A, B, C, D

(4) B, C, E

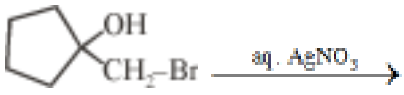
44) 0.1 gm of organic compound was analysed by Kjeldahl's method. In analysis produced NH_3 absorbed in 30 ml N/5 H_2SO_4. The remaining acid required 20 ml N/10 NaOH for neutralisation. Calculate percentage of nitrogen in organic compound :-

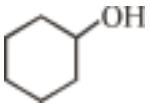
(1) 28 %

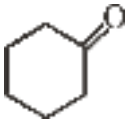
(2) 56 %

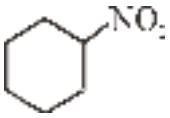
(3) 35 %

(4) 70 %

45)  major product

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Which of the following 3-different genera are placed in family solanaceae ?

(1) *Triticum*, *Mangifera* and *Petunia*

(2) *Solanum*, *Petunia* and *Datura*.

(3) *Solanum*, *Petunia* and *Mangifera*

(4) *Solanum*, *Pisum* and *Triticum*

2) A student observed a unicellular organism under a microscope that showed both autotrophic and hetrotrophic mode of nutrition. The organism produce food by chloroplast in the presence of sunlight, but in the absence of light it survives by predation. It lacked cell wall, but had a protein rich covering that provided flexibility. Based on the above observation the organism most appropriately belongs to which group?

(1) Cyanobacteria

(2) Kingdom plantae

(3) Euglenoid

(4) Slime mould

3) Which of the following statement is wrong ?

(1) Amoeboid protozoans live in fresh water, sea water and moist soil.

(2) The parasitic members of flagellated protozoans cause diseases such as sleeping sickness

(3) Ciliated protozoans are aquatic and actively moving organism because of the presence of thousands of cilia.

(4) Amoeboid protozoans have an infectious spore like stage in their life cycle.

4) Match column I with column II and choose the correct answer from the options given below :

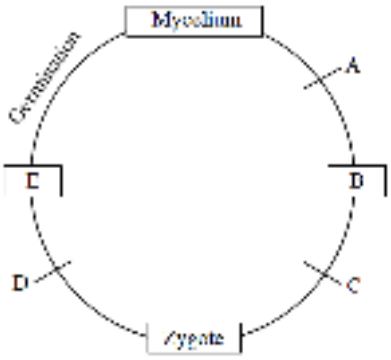
Column-I		Column-II	
a	Virus	(i)	Naked RNA
b	Viroid	(ii)	Protein only
c	Prion	(iii)	Capsid
d	Lichen	(iv)	Symbiotic association
		(v)	DNA or RNA
		(vi)	Cell wall

(1) a-i, iii; b-v; c-ii; d-vi

(2) a-v; b-i, vi; c-ii; d-iv

(3) a-iii, v; b-i; c-ii; d-iv, vi

(4) a-iii; b-v; c-i; d-iv, vi



5) The above diagram shows a generalize life cycle of a fungus. The appropriate terms for A to E are :-

	A	B	C	D	E
(1)	Hyphae	Mitosis	Meiosis	Fertilization	Spore
(2)	Spores	Meiosis	Karyogamy	Dikaryon	Plasmogamy
(3)	Hyphae	Meiosis	Spore	Mitosis	Fertilization
(4)	Plasmogamy	Dikaryon	Karyogamy	Meiosis	Spores

(1) 1

(2) 2

(3) 3

(4) 4

6) Match the column-I, II, III and select the correct one :-

	Column-I		Column-II		Column-III
a.	Bryophyta	(i)	Vascular cryptogams	(P)	*Lycopodium*
b.	Gymnosperm	(ii)	Amphibians of plant kingdom	(Q)	Moss

c.	Algae	(iii)	Successful land plant	(R)	*Volvox*
d.	Pteridophyta	(iv)	Aquatic nature	(S)	*Pinus*

(1) a-(ii)-Q, b-(iii)-S, c-(iv)-R, d-(i)-P

(2) a-(ii)-R, b-(iii)-P, c-(i)-Q, d-(iv)-S

(3) a-(iii)-P, b-(ii)-R, b-(iv)-S, d-(i)-Q

(4) a-(iii)-R, b-(ii)-P, c-(iv)-Q, d-(i)-S

7) **Assertion (A) :** *Selaginella* and *Salvinia* are heterosporous pteridophytes.
Reason (R) : Both genera produce two kinds of spores, macro (large) and micro (small) spores.

(1) (A) is correct but (R) is incorrect.

(2) (A) is incorrect but (R) is correct.

(3) (A) and (R) both are correct and (R) is the correct explanation of (A).

(4) (A) and (R) both are correct but (R) is not the correct explanation of (A).

8) Select ***incorrect*** statement from the following regarding gymnosperms :-

(1) Multicellular female gametophyte is retained within megasporangium.

(2) Pollen grains released from the microsporangium are carried in air currents to reach the ovules.

(3) The megaspore mother cell divides meiotically to form four megaspores.

(4) All four megaspores develops into a multicellular female gametophyte.

9) List of some of the genera is given below along with their codes :-
• *Hibiscus* – A
• *Solanum* – B
• *Allium* – C
• *Pisum* – D
Select the correct option in which codes of genera are arranged in ascending order on the basis of number of carpel(s) in their gynoecium :-

(1) D→B→A→C

(2) D→B→C→A

(3) A→B→D→C

(4) B→D→C→A

10) Match **Column-A** & **Column-B** to find out correct answer :-

Column-A		**Column-B**	
i.	Modified shoot	a	Peduncle
ii.	Racemose inflorescence	b	Basipetal arrangement
iii.	Cymose inflorescence	c	Flower
iv.	Axis of Inflorescence	d	Acropetal Arrangement

(1) i-a, ii-b, iii-c, iv-d

(2) i-b, ii-c, iii-d, iv-a

(3) i-c, ii-d, iii-b, iv-a

(4) i-d, ii-c, iii-a, iv-b

11) Identify the leaf shown below as well as select the plant of its occurrence and select the right option for the two together.



(1) Opposite, banana

(2) Pinnately compound, neem

(3) Palmately compound, silk cotton

(4) Alternate, guava

12) Pericycle in the form of semi-lunar patches of sclerenchyma is present in :

(1) Monocotyledonous stem

(2) Monocotyledonous root

(3) Dicotyledonous stem

(4) Dicotyledonous root

13) (A) In conjoint type of vascular bundles, the xylem and phloem are jointly situated along the same radius.
(B) All tissues on the inner side of endodermis constitute the stele.
(C) In dicot stems vascular bundles are closed type.
(D) In leaf of dicot plants the abaxial epidermis generally bears more stomata than adaxial epidermis.
Select the incorrect statement/s :-

(1) Only C

(2) B and C

(3) C and D

(4) A and C

14) Which of the following statement is **incorrect** regarding typical angiosperm anther ?

(1) Each cell of sporogenous tissue is capable of giving rise to a microspore tetrad.

(2) Microspore mother cell produces four microspore through the process of mitosis.

(3) Microsporangium is tetrasporangiate.

(4) Both 2 & 3

15)

The functional megaspore undergoes-

(1) 2 meiotic divisions to form mature embryo sac

(2) 3 mitotic divisions to form mature embryo sac

(3) 2 mitotic divisions to form mature embryo sac

(4) 4 meiotic divisions to form mature embryo sac

16) Consider the following statements and choose the correct option :-
(a) In flowering plants both types of gametes (male and female gametes) are motile and brought together for fertilization by pollination.
(b) If both male and female flowers are present on the same plant it prevents autogamy but not geitonogamy.
(c) Majority of wind pollinated flowers are large, colourful, fragrant and rich in nectar.
(d) If the female parent bears bisexual flowers, removal of pistil from the flower bud before anther dehisces. This step is referred to as emasculation.

(1) A, B and D

(2) B, C and D

(3) A and B

(4) Only B

17) Enzyme which catalyses joining of C–O, C–S, C–N, P–O etc. bonds are included in which of the following class ?

(1) Oxidoreductases

(2) Ligases

(3) Isomerases

(4) Lyases

18) Match List-I with List-II

List-I		**List-II**	
A.	Ribozyme	(i)	Enable glucose transport into cell
B.	Lecithin	(ii)	Nucleic acid behave like enzyme
C.	GLUT-4	(iii)	Belongs to phospholipid
D.	Vitamin	(iv)	Co enzyme

(1) A-iv, B-iii, C-ii, D-i

(2) A-ii, B-i, C-iii, D-iv

(3) A-ii, B-iii, C-i, D-iv

(4) A-iv, B-iii, C-i, D-ii

19) Which of the following is not strictly a macromolecule.

(1) Lipid

(2) Protein

(3) Nucleic acid

(4) All of the above

20) Read the following statements :
a. There is not a complete one-to-one overlap between the chl-a absorption spectrum and action spectrum of photosynthesis.
b. Chlorophyll b is yellow green in the chromatogram.
c. Reaction centre of PS II has absorption peak at 700 nm.
d. CO_2 is utilised for the carboxylation of RuBP during photorespiration.
e. Light is the major limiting factor for photosynthesis.
Choose the incorrect options :

(1) c, d & e

(2) a, b & d

(3) b, c & d

(4) a, d & e

21) Given below are two statements :
Statement-I : Carboxylation is the most crucial step of the calvin cycle where CO_2 is utilised for the carboxylation of RuBP.
Statement-II : Calvin cycle occurs in all photosynthetic plants ; it does not matter whether they have C_3 or C_4 (or any other) pathways.
In the light of the above statements, choose the most appropriate answer from the options given below -

(1) Both **statement-I** and **statement-II** are correct

(2) Both **statement-I** and **statement-II** are incorrect

(3) **Statement-I** is correct but **statement II** is incorrect.

(4) **Statement-I** is incorrect but **statement-II** is correct.

22) Given below are two statements one is labelled as Assertion (A) and the is labelled as Reason (R):-
Assertion (A): Aerobic respiration is the process that leads to a complete oxidation of organic substances in the presence of oxygen and release CO_2, water (H_2O) and a large amount of energy. This type of respiration is most common in higher organisms.
Reason (R):- During aerobic respiration O_2 is consumed and CO_2 is released. The ratio of the volume of CO_2 evolved to the volume of O_2 consumed in respiration is called the respiratory quotient (RQ).
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

23) Match column-I with column-II and choose the correct option :-

	Column-I		**Column-II**

(A)	Electron transport system	(i)	Terminal hydrogen acceptor
(B)	Citric acid	(ii)	Inner mitochondrial membrane
(C)	1-3 Bisphosphoglycerate	(iii)	Tricarboxylic acid cyclic
(D)	Oxygen	(iv)	Glycolysis

(1) A-ii, B-iv, C-iii, D-i

(2) A-iii, B-ii, C-iv, D-i

(3) A-ii, B-iii, C-iv, D-i

(4) A-i, B-ii, C-iii, D-iv

24) Which of the following is natural auxin and synthetic auxin (respectively) :-

(1) IAA, NAA and IBA, 2,4-D

(2) IAA, 2,4-D and NAA, IBA

(3) IAA, IBA and NAA, 2,4-D

(4) IBA, 2, 4- D and IAA, NAA

25) All changes that an organism goes through during it's life cycle from germination of seed to senescence are called together :-

(1) Growth

(2) Development

(3) Plasticity

(4) Senescence

26) Select correct statements :-
(A) Gibberellins are used to increase the length of grape stalks
(B) Kinetin does not occur naturally in plants
(C) ABA helps overcome the apical dominance
(D) Auxin initiates germination in peanut seeds

(1) A & B

(2) A, B & C

(3) A, B & D

(4) B & D

27) While studying the karyotype of a newly identified species, a researcher observed that male possess two similar sex chromosomes, while female possess two dissimilar sex chromosomes. further analysis revealed that female produce two type of gametes with respect to sex chromosomes while male produces only one type of gamete.
Which of the following sex determination mechanism is most likely operating in this organism :-

(1) XX - XY type

(2) XX - XO type

(3) ZW - ZZ type

(4) Haplodiplontic type

28) Match list-I with list-II

	List-I		List-II
(A)	Haemophilia	(I)	Autosome linked recessive
(B)	Thalassemia	(II)	Sex linked recessive
(C)	Down's syndrome	(III)	Absence of one of the X chromosome
(D)	Turner's syndrome	(IV)	Trisomy of 21 chromosome

Choose the correct answer from the option given below :-

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-I, B-III, C-IV, D-II

(4) A-II, B-I, C-IV, D-III

29) When Mendel self hybridised the F_1 plants having RrYy genotype he found that 3/4th of F_2 plants had yellow seeds and 1/4th had green seeds. Select the incorrect answer from given options -

(1) The yellow and green colour segregated in a 3 : 1 ratio

(2) The round and wrinkeled seed shape segregated in a 3 : 1 ratio.

(3) It is a dihybrid cross showing inheritance of two genes.

(4) It is a monohybrid cross showing inheritance of one gene.

30) Which symbol of pedigree is correctly matched ?

(1)  - Consanguineous mating

(2)  - Unaffected offspring

(3)  - Sex unspecified

(4)  - Unaffected female

31) Fill in the blank :
Alfred Sturtevant used the frequency of _____ between gene pairs on the same chromosome as a measure of the distance between genes.

(1) Sex chromosomes

(2) Linkage

(3) Independent assortment

(4) Recombination

32) Based on the extract of H.G. P. match the column-A with column-B

Column-A	Column-B

A.	Number of Gene on chromosome 1	1.	30000
B.	Number of gene on chromosome y	2.	< 2%
C.	Genome coding for protein	3.	2968
D.	Dissimilarity of nucleotide bases among individuals	4.	0.01%
		5.	99.9%
		6.	231
		7.	3000
		8	99.9%

(1) A-1, B-6, C-8, D-5

(2) A-3, B-6, C-5, D-7

(3) A-1, B-6, C-4, D-5

(4) A-3, B-6, C-2, D-4

33) In a double stranded DNA 1200 base pairs are present then find out the number of nucleosomes, Total N-Bases, Length of DNA (nm)

(1) 6, 2400, 360 nm

(2) 4, 2400, 360 nm

(3) 6, 1200, 408 nm

(4) 6, 2400, 408 nm

34) Choose incorrect statement :

(1) RNA polymerase binds to promoter and initiates transcription (initiation)

(2) There is single DNA dependent RNA polymerase that catalyses transcription of all types of RNA in bacteria.

(3) There are at least three RNA polymerase in the nucleus of prokaryotes.

(4) All three RNAs are needed to synthesize a protein in a cell.

35) During splicing process, In tailing adenylate residues are added at 3'-end of hnRNA ?

(1) In accordance to initiation-factor (σ)

(2) A template dependent manner.

(3) In accordance to termination-factor (ρ)

(4) A template independent manner.

36) On the template with polarity 3' -5' the replication is :-

(1) Continuous

(2) Semi continuous

(3) Discontinuous

(4) Semi discontinuous

37) Choose the incorrect match :-

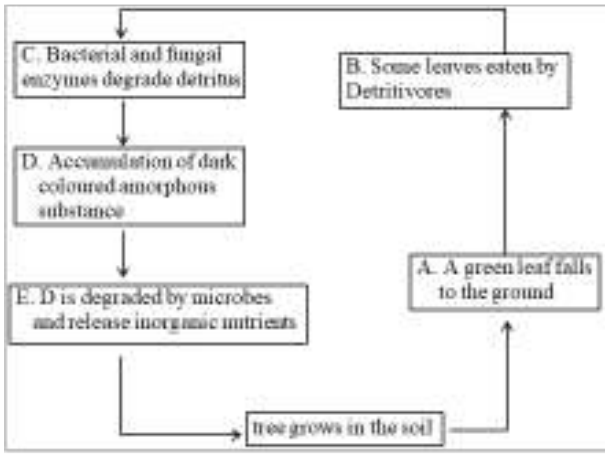
(1) AUG – Start codon
(2) UAA – Stop Codon
(3) UGG – Tryptophan
(4) GAG – Methionine

38) Given below are two statements one is labelled as Assertion and the other is labelled as Reason :
Assertion : The mediterranean orchid *Ophrys* employs "sexual deceit" to get pollination done by a species of bee.
Reason : The female wasp uses the fig fruit as an oviposition (egg-laying) site.
In the light of the above statement, choose the correct answer from the options given below

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

39) In 1981 the r-value for human population in India was :

(1) 0.015
(2) 0.12
(3) 0.0205
(4) 2.718

40) 

The above diagram shows decomposition cycle in terrestrial ecosystem. The most appropriate terms for A to E are :

	(1)	(2)	(3)	(4)
A	Mineralisation	Detritus	Fragmentation	Detritus
B	Humification	Fragmentation	Detritus	Fragmentation
C	Catabolism	Catabolism	Catabolism	Humification
D	Fragmentation	Humification	Humification	Catabolism
E	Detritus	Mineralisation	Mineralisation	Mineralisation

(1) 1

(2) 2

(3) 3

(4) 4

41) Given below are two statement-
Statement-I : Death of organism is the beginning of the detritus food chain/web.
Statement-II : In a terrestrial ecosystem, GFC is the major conduct for energy flow.
Choose most appropriate answer from given below options?

(1) Statement-I and Statement-II both are correct.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Statement-I is correct but statement-II is incorrect.

(4) Both Statement-I and Statement-II are incorrect.

42) Select the incorrect statement about ecological pyramid -

(1) The base of each pyramid represents the producer.

(2) The apex of each pyramid represents the top level consumer.

(3) It does not accommodate food web.

(4) Pyramid of energy is always inverted.

43) How many of the following methods are of in-situ and ex-situ conservation respectively ?
(A) National park
(B) Hotspot
(C) Sacred groves
(D) Cryopreservation
(E) Biosphere reserve
(F) Wild life safari park

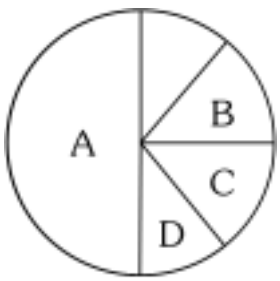
(1) 4, 2

(2) 2, 4

(3) 3, 3

(4) 5, 1

44)



Find out the A, B, C and D :

(1) A-Fishes, B-Mammals, C-Birds, D-Reptiles

(2) A-Mammals, B-Birds, C-Reptiles, D-Amphibians

(3) A-Birds, B-Fishes, C-Amphibians, D-Mammals

(4) A-Fishes, B-Birds, C-Reptiles, D-Amphibians

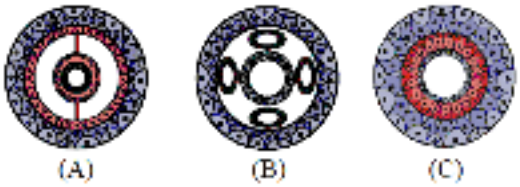
45) Which of the following is the cause of extinction of cichlid fish :-

(1) Over exploitation
(2) Introduction of Nile perch in lake victoria
(3) Loss of Habitat
(4) Extinction of Host

46) How many malpighian tubules are present in Cockroach?

(1) 60-100
(2) 90-110
(3) 100-150
(4) 70-90

47) Identify the following structure A, B and C. These structures respectively present in :-



(1) Annelida, Aschelminthes, Platyhelminthes
(2) Aschelminthes, Annelida, Platyhelminthes
(3) Platyhelminthes, Aschelminthes, Annelida
(4) Aschelminthes, Platyhelminthes, Annelida

48) Which of the following character is not found in all chordates :-

(1) True coelom
(2) Dorsal nerve cord
(3) Pharyngeal gill clefts
(4) Diaphragm

49) Which of the following features are present in chordates ?

(1) Dorsal heart, post anal tail, dorsal central nervous system
(2) Ventral heart, post anal tail, gill slits
(3) Dorsal heart, pharynx perforated by gill slits, dorsal central nervous system
(4) Ventral heart, hollow notochord, post anal tail

50) **Statement I :** In sponges water enters through minute pores (ostia) in the body wall into a central cavity (spongocoel)
Statement II : The body is supported by a skeleton made up of spicules or spongin fibre.

(1) Both statement-I and II are correct.

(2) Only statement-I is correct.

(3) Only statement-II is correct.

(4) Both statement-I and II are incorrect.

51) **Assertion :** *Trygon* cannot maintain their internal temperature.
Reason : Poison sting present in *Trygon* helps in temperature regulation.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

52) Read the following statements and choose the option with correct statements.

(1) Blood cells secrete collagen fibres

(2) Osteocytes are present in lacuna

(3) Chondrocytes are present in lacuna

(4) Both (2) & (3)

53) Choose correct statements ?
(a) The most common species of the frog found in India is Rana tigrina.
(b) Dorsal side of frog is olive green.
(c) Copulatory pad and vocal sacs are found in both male and female.
(d) Winter sleep is known as aestivation.

(1) (a) & (b)

(2) (a) & (c)

(3) Only (a)

(4) (a) & (d)

54) Read column I, II and III and identify their correct match :

	Column I		**Column II**		**Column III**
(A)	Fishes	(P)	Uric acid	(i)	Single circulation
(B)	Birds	(Q)	2-chambered heart	(ii)	Double circulation
(C)	Ureotelic	(R)	4-chambered heart	(iii)	Amphibia
(D)	Uricotelic	(S)	Urea	(iv)	Lizard

(1) A-Q-iii, B-R-iv, C-S-ii, D-P-i

(2) A-Q-i, B-R-ii, C-S-iii, D-P-iv

(3) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(4) A-Q-ii, B-P-i, C-S-iv, D-R-iii

55) In which of the following chromosome, the centromere is situated close to its end forming one extremely short and one very long arm?

(1) Metacentric

(2) Sub-metacentric

(3) Telocentric

(4) Acrocentric

56) Given below are the two statements:
Statement-I: Proteins synthesised by ribromes on endoplasmic reticulum are modified in the cisternae of golgi-apparatus before they released from its trans face.
Statement-II: Larger and more numerous nucleoli are present in cells activity carrying out protein synthesis.

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

57) Match Column-I with Column-II and choose the correct options?

Column-I		**Column-II**	
(a)	Endoplasmic reticulum	(I)	Tonoplast
(b)	Golgi apparatus	(II)	Synthesis of steroidal hormone.
(c)	Lysosomes	(III)	Synthesis of glycoprotein.
(d)	Vacuole	(IV)	Hydrolytic enzymes.

(1) a-II, b-III, c-IV, d-I

(2) a-III, b-IV, c-I, d-II

(3) a-IV, b-III, c-II, d-I

(4) a-III, b-II, c-IV, d-I

58) **Assertion :** Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis.
Reason : The first two stages of prophase I are relatively short-lived compared to the next stage that is pachytene.

(1) Both assertion and reason are true but reason is not the correct explanation of assertion.

(2) Both assertion and reason are true and reason is the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

59) Read the following statements and choose the correct option.
(a) Spindle fibres attach to kinetochores of chromosomes during prophase.
(b) Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.
(c) Centromeres split and chromatids separate during anaphase
(d) Chromatids move to opposite poles during telophase
(e) Nucleolus, golgi complex and ER reform during metaphase.

(1) a, b and c are correct

(2) c and e are correct

(3) b and c are correct

(4) a and b are correct

60) Endothelium of blood vessels is made up of :

(1) Simple cubiodal epithelium

(2) Simple columnar epithelium

(3) Simple squamous epithelium

(4) Urothelium

61) **Assertion :** The cardiac output of an athlete will be much higher than that of an ordinary man.
Reason : The body has the ability to alter the stroke volume as well as the heart rate and there by the cardiac - output.

(1) Assertion is true but Reason is false.

(2) Both Assertion and Reason are correct and Reason is not the correct explanation of Assertion.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

62) A person breathes in some volume of air by forced inspiration after having a forced expiration. This quantity of air taken in is called :

(1) Total lung capacity

(2) Tidal volume

(3) Vital capacity

(4) Inspiratory capacity

63) Choose the incorrect statements.

(1) Alveoli are the primary sites of exchange of gases.

(2) RBCs contain a very low concentration of the enzyme carbonic anhydrase.

(3) Pneumotaxic centre can reduce the duration of inspiration

(4) Oxygen dissociation curve is sigmoid.

64) Match the columns :-

1.	Vasa recta	A.	Release when GFR fall
2.	Renin	B	25-30 gm/day
3.	Urea excreted	C.	Brush border cuboidal epithelium
4.	PCT	D.	Peritubular capillary parallel to loop of henle

(1) 1-D, 2-A, 3-B, 4-C

(2) 1-A, 2-B, 3-C, 4-D

(3) 1-C, 2-B, 3-A, 4-D

(4) 1-B, 2-A, 3-D, 4-C

65) Which of the following is not a function of kidney ?

(1) Secretion of albumin

(2) Regulation of acidity of body fluids

(3) Removal of urea

(4) Regulation of BP

66) Resting membrane potential is maintained by :-

(1) Positively charged protein in axoplasm

(2) Na^+ – K^+ pump

(3) Passive diffusion of Na^+ by ion channels

(4) All of the above

67) **Statement-I :** In human body the neural system and the endocrine system jointly coordinate & integrate all the activities of the organs.
Statement-II : The neural system provides an organised network of point to point connection for a quick cordination

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

68) Fall in blood calcium levels stimulates the release of _____ from the _____ gland.

(1) Thyroxine; thyroid

(2) Calcitonin; parathyroid

(3) Parathormone; parathyroid

(4) Insulin; pancreas

69) Select the incorrect Statement -

(1) JG cells of kidney produce a peptide hormone called erythropoietin.

(2) Androgen acts on the peripheral nervous system and influence the male sexual behaviour.

(3) GIP inhibits gastric secretion and motility.

(4) PTH is a hyper calcemic hormone.

70) Match the following columns & choose the correct option :

Hormone	Function

A.	Adrenaline	(i)	BMR Regulation
B.	Thyroxine	(ii)	Emergency response
C.	Melatonin	(iii)	Immunosuppressive
D.	Cortisol	(iv)	Sleep-wake cycle

Options:

(1) A-ii, B-i, C-iv, D-iii

(2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-iii, C-ii, D-i

(4) A-iii, B-ii, C-iv, D-i

71) **Statement-I :-** Pancreas is considered as mixed gland because it performs both endocrine as well as exocrine function.
Statement-II :- Thymus and ovary function only as endocrine glands and do not have any role related to cell production or maturation in the human body.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

72)

According to the sliding filament theory, during muscle contraction :

(1) Actin and myosin filaments slide over each other to increase the length of the sarcomere.

(2) Length of A–band does not change.

(3) I–band increases in length

(4) The actin filaments slide away from A–band resulting in shortening of sarcomere.

73) The vertebral column is differentiated into X(7), Y(12), Z(5) regions starting from the skull here 7, 12, 5 are the number of vertebrae present in X,Y,Z region. Choose the correct about X, Y, Z.

	X	**Y**	**Z**
(1)	Lumber	Thoracic	Saccral
(2)	Cervical	Thoracic	Coccygeal
(3)	Cervical	Thoracic	Lumber
(4)	Cervical	Thoracic	Saccral

(1) 1

(2) 2

(3) 3

(4) 4

74) By the end of 12 weeks, most of the major organs system are formed, for example, limbs and _____ are well developed.

(1) External genital organs

(2) Eyelashes

(3) Hair on head

(4) Eyelids

75) Read the following statements A-E :
A- Placenta acts as an exocrine gland.
B-Placenta is formed by the interdigitation of chorionic villi of developing embryo and maternal tissue.
C-hCG, hPL and relaxin are produced only during pregnancy.
D-The placenta is connected to the embryo through spermatic cord.
E-During pregnancy the levels of other hormones like estrogen, cortisol, prolactin etc. are increased several folds in the maternal blood.
Select the option which includes the incorrect statements only.

(1) A, B, C

(2) A and D only

(3) B, C, E

(4) D and E only

76) Match the list-I with list-II and select correct answer about them

List-I		**List-II**	
a.	Ampulla of fallopian tube	i.	First movement of foetus
b.	Zona pellucida	ii.	8-16 blastomere stage embryo
c.	Morula	iii.	Acts as an endocrine tissue
d.	Placenta	iv.	Block the entry of additional sperms into the ovum
e.	During fifth month of pregnancy	v.	The site where fertilisation take place

(1) a-i, b-ii, c-iii, d-iv, e-v

(2) a-v, b-iv, c-ii, d-iii, e-i

(3) a-iv, b-iii, c-v, d-i, e-ii

(4) a-iii, b-i, c-iv, d-ii, e-v

77) Read the following two statements :
Statement-I : The placenta is connected to the embryo through an umbilical cord which help in transport of substances to and from the embryo.
Statement-II : In the later phase of pregnancy a hormone called cortisol is also secreted by the ovary.
On the light of above statements choose the correct answer from the options given below :

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

78) Find out incorrect pair :

(1) Colostrum - Milk produced during initial few days of lactation.

(2) Gestation period - Average duration of human pregnancy is about 9 month.

(3) Signal for parturition - Originate from fully developed foetus and pituitary gland of mother.

(4) End of 12 week - Most of major organ system are formed.

79) Which of the following statement is not matched ?

1.	Veneral disease	Trichomoniasis
2.	Artificial Insemination	Done in very low sperm count
3.	Medical termination of pregnancy	Safe if done during the first trimester
4.	Genital herpes	Completely curable if detected early

(1) 1

(2) 2

(3) 3

(4) 4

80) Read the following statements:
A - Health is affected by genetic disorders, Infections and life style.
B - Among non-infectious diseases, AIDS is the major cause of death.
C - Ringworm is caused by Trichophyton
D - The primary and secondary immune responses are carried out only with the help of T-lymphocytes.
E - The cannabinoids are known for their effect on respiratory system of the body.
Select the option which includes incorrect statements only :

(1) A,B,C,D

(2) B,C,D,E

(3) B,D,E

(4) A,B,C

81) **Assertion (A) :** When our body encounters a pathogen for the first time it produces a response called primary response which is of low intensity.
Reason (R) : Subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

82) Which of the following statement is correct?

(1) Drug and alcohol abuse do not affect our health adversely.
(2) Yoga has been practised since time immemorial to achieve physical and mental health.
(3) Health decreases longevity of people.
(4) Health would be defined at the state of physical, mental and sexual well being.

83) Which of the following set of cells are affected by HIV?

(1) Macrophage and B-lymphocytes
(2) Macrophage and T-lymphocytes
(3) Eosinophils and mast cells
(4) T-lymphocytes and B-lymphocytes

84) Match the column-I with column-II.

Column-I		**Column-II**	
(i)	Thomas malthus	(a)	Use and disuse of organ
(ii)	Lamarck	(b)	Theory of natural selection
(iii)	Hugo de vries	(c)	Principles of population
(iv)	Darwin	(d)	Mutation theory

(1) i-a, ii-c, iii-d, iv-d
(2) i-c, ii-a, iii-d, iv-b
(3) i-a, ii-c, iii-d, iv-b
(4) i-c, ii-a, iii-b, iv-d

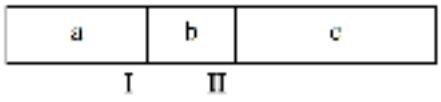
85) Name of the dinosaur, who was about 20 feet in height and had huge fearsome dagger like teeth.

(1) Tyrannosaurs rex
(2) Triceratops
(3) Brachiosaurus
(4) Stegosaurus

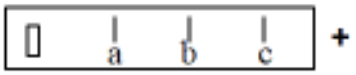
86) The processes, which are collectively referred to as downstream processing :-

(1) Separation and production
(2) Purification and expression
(3) Separation and purification
(4) Selection of recombinants

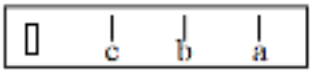
87) The given segment of DNA has restriction sites I and II, which creates restriction fragments a, b and c.

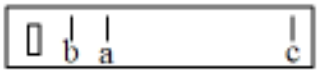


Select the correct diagram of gel.

(1) 

(2) 

(3) 

(4) 

88) Which of the following statements are correct regarding the process of RNA interference-
(i) The method involves silencing of a specific mRNA due to a complementary dsRNA molecules.
(ii) This is used to prevent the infection of protozoans
(iii) It takes place in some eukaryotic and all prokaryotic organisms as a method of cellular defense.
(iv) It is novel strategy to produce pest resistant plants.

(1) (i), (ii) and (iv)

(2) (ii) and (iii)

(3) (i) and (iv)

(4) (i), (ii) and (iii)

89) Fill in the blanks
" _____is not present in the mature insulin and is removed during maturation into insulin"

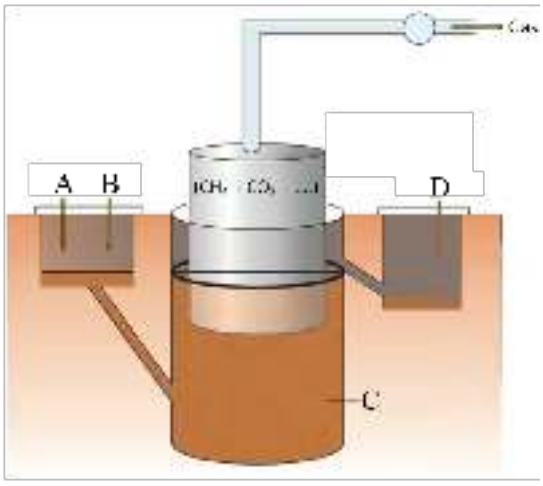
(1) C peptide

(2) A Peptide

(3) B peptide

(4) A & B peptide

90) Which of the following is correct labelling of C in the given diagram.



(1) Aerobic bacteria
(2) Sludge
(3) Digester
(4) gas holder

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	3	3	3	4	3	3	1	2	2	2	2	1	4	1	1	2	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	2	2	4	2	4	3	3	4	3	4	1	4	2	3	4	2	4	1
Q.	41	42	43	44	45															
A.	2	2	3	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	2	3	1	4	2	2	2	1	1	3	1	1	2	4	4	3	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	1	2	1	3	3	4	3	2	3	4	2	4	3	3	4	2	2	1
Q.	86	87	88	89	90															
A.	4	4	2	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	4	3	4	1	3	4	2	3	3	3	1	4	2	4	2	3	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	3	3	2	1	3	4	4	3	4	4	4	3	4	1	4	2	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	1	4	2	3	1	4	2	1	3	4	1	2	4	1	1	1	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	2	1	1	2	1	3	2	1	3	2	3	1	2	2	3	3	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	2	2	1	3	2	3	1	3										

SOLUTIONS

PHYSICS

1) $v^2 = t \Rightarrow v = t^{1/2} \Rightarrow a = \frac{dv}{dt} = \frac{1}{2}t^{-1/2}$

$\text{At } t = 4s,\ a = \frac{1}{2}(4)^{-1/2} = \frac{1}{4}\,m/s^2$

2) $\sum u = 0$
$y = 3x + 5x + 7x$
$\frac{y}{x} = \frac{15}{1}$

3) $D = (4.23 \pm 0.01)$ cm
$d = (3.89 \pm 0.01)$ cm
$\Delta t = (D - d)/2 = [(4.23 \pm 0.01) - (3.89 \pm 0.01)]/2$
$= [(4.23 - 3.89) \pm (0.01 + 0.01)]/2$
$= (0.34 \pm 0.02)/2$ cm $= (0.17 \pm 0.01)$ cm

4)

(A) $E = \frac{1}{2}LI^2 \Rightarrow [L] = \frac{[E]}{[I^2]}$

$\Rightarrow [L] = \frac{\left[M^1L^2T^{-2}\right]}{A^2} = \left[M^1L^2T^{-2}A^{-2}\right]$ $(A \rightarrow R)$

(B) $E = \frac{1}{2}KT$

$\Rightarrow [K] = \frac{[E]}{[T]} = \frac{\left[M^1L^2T^{-2}\right]}{[K]} = \left[M^1L^2T^{-2}K^{-1}\right]$ $(B \rightarrow D)$

(C) $\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow [\mu_0] = \frac{[F]}{[I^2]} = \frac{\left[M^1L^1T^{-2}\right]}{A^2}$

$[\mu_0] = [M^1L^1T^{-2}A^{-2}]$ $(C \rightarrow S)$

(D) $R = \rho\frac{\ell}{A} = [\rho] = [RL] = \left[\frac{\rho}{I^2}.L\right]$

$\Rightarrow [\rho] = \frac{\left[M^1L^2T^{-3}L\right]}{A^2} = \left[M^1L^3T^{-3}A^{-2}\right]$ $(D \rightarrow Q)$

5) $m_2g - T = m_2a$
$T - \mu k m_1 g = m_1 a$

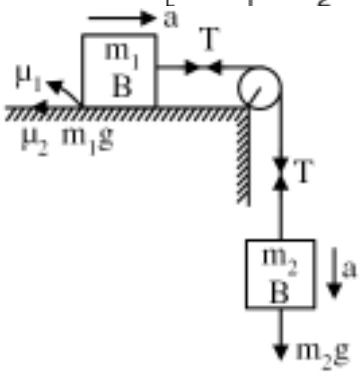
$$\Rightarrow a = \frac{(m_2 - \mu km)\, g}{m_1 + m_2}$$

For the block of mass 'm_2'

$$m_2 g - T = m_2 \left[\frac{m - 2 - \mu_k}{m_1 + m_2}\right]$$

$$m_2 g - T = m_2 \left[\frac{m_2 - \mu_K m - 1}{m_1 + m_2}\right]$$

$$m_2 g = m_2 g \left[\frac{m_2 - \mu_K m - 1}{m_1 + M_2}\right]$$



6) momentum acquired = Area of force-time graph

$= \frac{1}{2} \times (2) \times (10) + 4 \times 10 = 10 + 40 = 50$ N-S

7) C.O.M.E $V_B \rightarrow v$

$V_A \rightarrow 2v$

loss in potential energy = gain in K.E.

$mg\ell + mg2\ell = \frac{1}{2}mV_B^2 + \frac{1}{2}m(2V_A)^2$

$3mg\ell = \frac{1}{2}m(5V^2)$

$\frac{6}{5}g\ell = V^2$

$V = \sqrt{\frac{6}{5}g\ell}$

8) $U = \frac{f}{2}\mu RT$

$U = \frac{5}{2}(3RT) + \frac{3}{2}(5RT) = 15RT$

9) Heat loss = Heat gain

$m_B S_w(80 - 50) = m_A S_W(50 - 30)$

$\frac{m_A}{m_B} = \frac{30}{20} = \frac{3}{2}$ ...(i)

$m_A + m_B = 40$ ...(ii)

solving equation (i) & (ii) $m_A = 24$ kg, $m_B = 16$ kg

10) $\because \left(\frac{\Delta Q}{\Delta t}\right)_A = \left(\frac{\Delta Q}{\Delta t}\right)_B$

$\therefore \frac{K_A A(100-70)}{30} = \frac{K_B A(70-35)}{70}$

$\Rightarrow K_A = \frac{K_B}{2} \Rightarrow \frac{K_A}{K_B} = \frac{1}{2}$

11) In the curves 1 - 2 and 3 - 4, we find that the pressure is directly proportional to temperature. So, the volume remains unchanged, i.e., gas does not work. The work done during the isobaric processes
2 - 3 and 1 - 4 are as follows :

$W_{2 \rightarrow 3} = P_2(V_3 - V_2)$

$W_{1 \rightarrow 4} = P_1(V_1 - V_4)$

Total work done = $P_2(V_3 - V_2) + P_1(V_1 - V_4)$

$\therefore W_T = P_2V_3 - P_2V_2 + P_1V_1 - P_1V_4$

Three moles has been given, so

$PV = nRT = 3RT$

$\therefore W_T = 3RT_3 - 3RT_2 + 3RT_1 - 3RT_4$

$= 3R(T_1 + T_3 - T_2 - T_4)$

$= 3R[T_1 + T_3 + T_2 - T_4]$

$= 3R[400 + 2400 - 800 - 1200]$

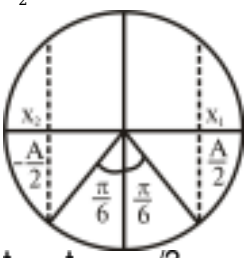
$= 3R \times 800 = 20 \times 10^3$ J = 20 kJ

12) $e = \frac{L_2 - 3L_1}{2} = \frac{32-30}{2} = 1$ cm

13)

$x_1 = 6.5 = \frac{A}{2} = A \sin \omega t_1$

$x_2 = -6.5 = -\frac{A}{2} = A \sin \omega t_2$



$\frac{t_2 - t_1}{T} = \frac{\pi/3}{2\pi}$

$(t_2 - t_1) = T \times \frac{1}{6} = \frac{12}{6} = 2$ sec

14) At equilibrium :

$mg \sin\theta = kA$

$\Rightarrow \quad A = \frac{mg \sin\theta}{K}$

As mg sin θ is constant, the time period for spring block system is given as

$T = 2\pi\sqrt{\frac{m}{k}}$

15) $E_x = -\frac{\partial V}{\partial x} = -(-5)$

$= 5; E_y = -\frac{\partial V}{\partial y} = -3$

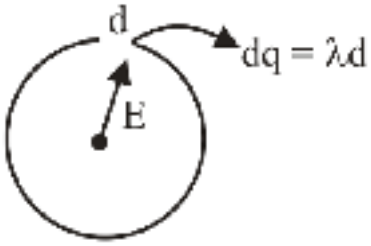
and $E_z = -\frac{\partial V}{\partial z} = -\sqrt{15}$

$E_{net} = \sqrt{E_x^2 + E_y^2 + E_z^2}$

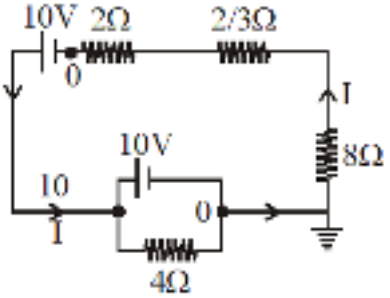
$= \sqrt{(5)^2 + (-3)^2 + \left(-\sqrt{15}\right)^2} = 7$

16)

By symmetry, $E_{centre} = \frac{Kdq}{R^2}$



$E = \frac{K(Q)d}{(2\pi R)(R^2)} \Rightarrow E \propto \frac{1}{R^3}$



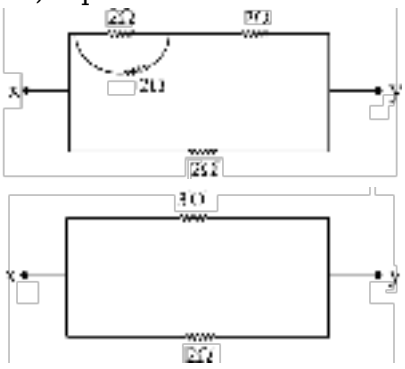
17)

no potential difference across 8Ω ⇒ i = 0

Potential difference across 4Ω is 10V

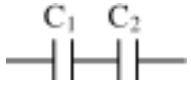
$\Rightarrow i = \frac{10}{4} = 2.5A$

18) Equivalent ckt.



$$R_{xy} = \frac{3 \times 2}{5} = \frac{6}{5}\text{ohm.}$$

19) By changing thickness of wire resistance of wire changes but ratio of $\frac{R_{AC}}{R_{CB}}$ does not change.

20) 

$C_{eq} = \frac{C_1C_2}{C_1 + C_2} = \frac{2 \times 4}{6} = \frac{8}{6} = \frac{4}{3}$

$Q_1 = C_1V_1 = 12$ mC

$Q_2 = C_2V_2 = 8$ mC

Combination can withstand charge 8 mC in series

$V_{combination} = \frac{8 \times 10^{-3}C \times 3}{4 \times 10^{-6}}$

$V_{combination} = 6$ KV

So it can withstand = 5 KV

21)

The magnetic field at the center due to a single ring of radius R is $B = \frac{\mu_0 i}{2R}$

The direction of magnetic field for all rings is the same (out of the page) as the current flows in the same direction in each ring.

$$B_{net} = \frac{\mu_0 i}{2r} + \frac{\mu_0 i}{2(2r)} + \frac{\mu_0 i}{2(2^2 r)} \ldots\ldots\infty$$

$$B_{net} = \frac{\mu_0 i}{2r}\left[1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \ldots\ldots\ldots\infty\right]$$

The term in the bracket in an infinite geometric progression with first term a = 1 and common ratio $r' = -\frac{1}{2}$

Sum of infinite GP $S_\infty = \frac{a}{1 - r'}$

$$S_\infty = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$$

$$B_{net} = \frac{\mu_0 I}{2r} \times \frac{2}{3}$$

$$B_{net} = \frac{\mu_0 i}{3r}$$

22)

E = (Area of loop) × (Volume) × Frequency × Total time

$E = 6 \times 10^4$ J

23) Current through capacitor,

$$I = \frac{E}{X_C} = \frac{E}{\frac{1}{\omega C}} = \omega CE = 2\pi\upsilon\, CE \text{ or } I \propto \upsilon$$

 Decrease in frequency υ of ac source decreases the conduction current. As displacement current is equal to conduction current, decrease in υ decreases displacement current in circuit.

24)

$m_1 \times m_2 = 1$

$$\frac{h_1}{h_0} \times \frac{h_1'}{h_0} = 1$$

$$\Rightarrow \frac{4}{h_0} \times \frac{2}{h_0} = 1$$

$$\Rightarrow h_0^2 = 8 \text{ cm}^2$$

$$\Rightarrow h_0 = 2\sqrt{2} \text{ cm}$$

25) $$\delta_1 = \left(\mu_{g-1}\right) A = \left(\frac{3}{2} - 1\right) A = \frac{A}{2}$$

$$\delta_2 = \left(w\mu_{g-1}\right) A = \left(\frac{3/2}{4/3} - 1\right) A = \frac{A}{8}$$

$$\Rightarrow \frac{\delta_1}{\delta_2} = \frac{A}{2} \times \frac{8}{A} = \frac{4}{1}$$

26) $$\frac{1}{f + 20} - \frac{1}{-(f + 20)} = \frac{1}{f} \Rightarrow f = 20\text{cm}$$

27)

• Fringe width remains unchanged since λ, D and d are constant
• Moving the source closer to S, introduces an initial path difference causing the central bright fringe (zero path difference) to shift downwards leading to a downward shift of the entire fringe pattern.

28) $$I = \frac{I_0}{2} \times \cos^2\left(30^0\right) \times \cos^2\left(30^0\right)$$

$$= \frac{I_0}{2} \times \frac{3}{4} \times \frac{3}{4}$$

$$= \frac{9I_0}{32}$$

29)

Current in zener diode

$$= \frac{12 - 8}{500} - \frac{8}{5 \times 10^3} = \frac{4}{500} - \frac{8}{5000}$$

= (8 – 1.6) mA = 6.4 mA

 Power lost in zener diode

= VI = 8 × 6.4 mW = 51.2 mW

30)

Applying forward bias reduces the potential barrier and narrows the depletion region, allowing charge carriers to flow across the junction.

A. When a forward bias is applied, the external voltage reduces the built-in potential barrier, making it easier for charge carriers to cross.
B. As the potential barrier decreases, the width of the depletion region (x) also decreases, allowing current to flow.

31) Total energy of SHM (After impulse)

$= m\omega^2A^2 = \frac{1}{2}m\omega^2A_1^2$

where A_1 is new amplitude

$\therefore A_1 = \sqrt{2}A$

32)

Standard wave equation is

$y=A \sin(\omega t - k x)$

$\omega = 100$ rad/s

$k = 1/10^{-1}$

Wave velocity

$v=\omega /k$

$=100 /(1/10)$

$=1000$ m/s

33)

Speed of sound, $V = \sqrt{\frac{\gamma RT}{M_w}}$

T → absolute temperature

M_w → Molecular mass

34) Elastic potential energy (U)

$U \propto \left(\frac{\Delta \ell}{L}\right)^2$

$\frac{U_1}{U_2} = \frac{\left(\frac{\ell}{L}\right)^2}{\left(\frac{2\ell}{2L}\right)^2} = 1$

35)

Excess pressure, radius of curvature.

$$r = \frac{r_1 r_2}{r_2 - r_1}$$

$$r = \frac{4 \times 5}{5-4} = \frac{20}{1} = 20\text{cm}$$

36) From COME,

$$\frac{1}{2}mV^2 = \frac{mgh}{\left(1 + \frac{h}{R}\right)}$$

$$\frac{1}{2}m\left(\frac{v_e}{2}\right)^2 = \frac{mghR}{h+R}$$

$$\frac{1}{8} \times (2gR) = \frac{ghR}{(h+R)}$$

$$\frac{h}{R+h} = \frac{1}{4} \Rightarrow \boxed{h = \frac{R}{3}}$$

37) Potential at the center is $V_{center} = -\frac{3GM}{2R}$

Potential at a distance r inside the sphere (r < R) is $V_r = -\frac{GM}{2R^3}(3R^2 - r^2)$

Potential at a distance r outside the sphere (r ⩾R) is $V_r = -\frac{GM}{r}$

Condition: $V_r = \frac{1}{2}V_{center} = \frac{1}{2}\left(-\frac{3GM}{2R}\right) = -\frac{3GM}{4R}$

Since $|V_r| < |V_{surface}|(\text{where } V_{surface} = -\frac{GM}{R})$,the point must be outside the sphere.

$$-\frac{GM}{r} = -\frac{3GM}{4R}$$

$$r = \frac{4R}{3}$$

38) $\frac{MR^2}{2} = \frac{ML^2}{12} + \frac{MR^2}{4}$

$L = \sqrt{3}R$

39) $T = \frac{2\pi r}{V_0}$ and $V^0 = \sqrt{\frac{GM}{r}}$

$$V_0^2 = \frac{GM}{r} \Rightarrow r = \frac{GM}{V_0^2}$$

$$T = \frac{2\pi}{V_0}\left(\frac{GM}{V_0^2}\right) = \frac{2\pi GM}{V_0^3}$$

40) $E_{peak} = c\, B_{peak}$

$= 3 \times 10^8 \times 2 \times 4\pi \times 10^{-7}$

$= 753$ volt/m

$$\hat{S} = \frac{1}{\mu_0}(\hat{E} \times \hat{B})$$

Since $\vec{B}$ is in $-\hat{i}$ & wave is travelling towards $\hat{k}$,

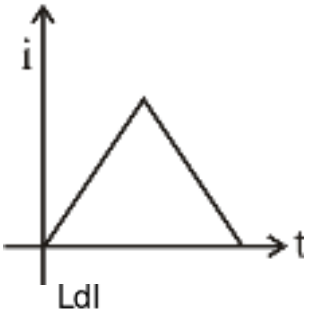
- $\vec{E}$ is towards $\hat{j}$
- Equation of electric field

$\vec{E} = 753 \cos(\omega t - kz)\hat{j}$

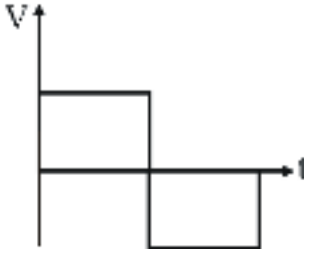
41)

For an inductor, induced voltage $e = L\frac{di}{dt}$.

When current increases linearly with time, $\frac{di}{dt}$ is constant and positive, so voltage is constant and positive.

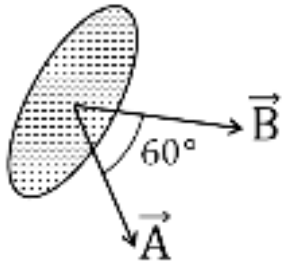


$e = \frac{LdI}{dt} = L \times$ (slope of i-t graph)



When current decreases linearly with time, di/dt is constant and negative, so voltage is constant and negative.

Thus the voltage–time graph consists of a constant positive value followed by a constant negative value, forming a rectangular (step-like) waveform.



42)

$\phi = BA\cos\theta$

$\phi = 4 \times \frac{1}{2} \times \frac{1}{2} = 1$ weber

43) $(mv)\hat{i} = \frac{mv}{\sqrt{3}}\hat{j} + m\vec{v}_2$

$$\vec{v}_2 = v\hat{i} - \frac{v}{\sqrt{3}}\hat{j} \quad |\vec{v}_2| = \frac{2}{\sqrt{3}}v$$

44) $$\eta = \frac{P_{out}}{P_{Input}}$$

$$\frac{88}{100} = \frac{e_s I_s}{e_p I_p} = \frac{P_{out}}{e_p I_p}$$

$$\frac{88}{100} = \frac{880}{2200 \times I_p}$$

$$I_p = \frac{10}{22} = 0.45 \text{ A}$$

45) $$B_c = \frac{\mu_0}{5\pi}\frac{I}{R}(\theta_R)$$

$$B = \frac{\mu_0 I}{4\pi r}(\sin\theta_1 + \sin\theta_2)$$

CHEMISTRY

46)

$NF_3 \rightarrow sp^3$, 1 lp, Pyramidal
$NO_3^- \rightarrow sp^2$, 0 lp, Trigonal planar
$BF_3 \rightarrow sp^2$, 0 lp, Trigonal planar
$H_3O^+ \rightarrow sp^3$, 1 lp, Pyramidal
$HN_3 \rightarrow sp$, Linear

47)

Explain Question : Count total antibonding electrons for O_2^-, O_2, and O_2^{2-} respectively.
Concept : This question is based on Molecular Orbital Theory
Solution :
O_2 ($16e^-$) has 6 antibonding electrons (ABMO). O_2^- ($17e^-$) adds one electron to an ABMO, for a total of 7. O_2^{2-} ($18e^-$) adds two, for a total of 8.
O_2 : (16 electron)- $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \sigma2pz^2, \pi2px^2 = \pi2py^2, \pi^*2px^1 = \pi^*2py^1$
O_2^- : (17 electron)- $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \sigma2pz^2, \pi2px^2 = \pi2py^2, \pi^*2px^2 = \pi^*2py^1$
O_2^{2-} : (18 electron)- $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \sigma2pz^2, \pi2px^2 = \pi2py^2, \pi^*2px^2 = \pi^*2py^2$
Final answer : 1

48)

Solution:
In –O–H···N–H in this case hydrogen bond is present between hydrogen and nitrogen and since nitrogen is less electronegative then oxygen so nitrogen is better lone pair donor so making more stronger hydrogen bond.
Final Answer: Option (2)

49) **Question Explanation :**
Question identify incorrect order dipole moment.
Concept :
This question is based on Dipole moment.
Solution :
The correct order for the dipole moment of chloromethanes is $CH_3Cl > CH_2Cl_2 > CHCl_3$. The vector addition of bond dipoles in these tetrahedral molecules leads to this specific trend, making option 3 incorrect.
Final Answer : Option (3)

50)

Explanation the Question:
Identify the incorrectly matched systematic and official names of an element.
Concept:
This question is based on IUPAC Nomenclature
Solution:
Unununnium is the systematic name for element 111, whose official name is Roentgenium (Rg). Darmstadtium (Ds) is element 110. Therefore, this match is incorrect.
Final Answer: (1)

51) 5d>3d>4d- due to Lanthanoid contraction

52) $Xe[\overset{+5}{Pt}F_6]$
Xenonhexafluoroplatinate (V)

53)

Explain Question: Asking about electronic configuration of d^5 in octahedral weak-field complex.

Concept: If pairing energy (P) > Δ_0, high-spin configuration occurs; electrons occupy orbitals to maximize unpaired electrons.
Solution: d^5 octahedral, weak-field ligand → high-spin → maximize unpaired electrons.
Example : $[Mn(H_2O)_6]^{2+}$ has d^5 configuration and weak-field H_2O ligands → high-spin $t_{2g}^3e_g^2$.
Final answer: Option (2)

54)

As per concept

55)

As per concept

56)

V_2O_5 → Used in contact process
$TiCl_4 + Al(CH_3)_3$ → Polymerisation of ethylene.
$PdCl_2$ → Oxidation of ethyne to ethanal
Nickel complex → Polymerisation of alkynes.

57)

Only Cl^- gives chromyl chloride test not covalent molecule

58) **Explain Question:** Asking about correct and incorrect statement?
Concept: Covalent compound, lonization enthalpy.
Solution:
Statement I: Correct – Boron forms only covalent compounds. (eg., BF_3, BCl_3, etc).
Statement II: Correct – Boron's first three ionization energies are very high, making it difficult to form B^{3+} ions.
since high ionization energy prevents ionic bond formation, **Statement II correctly explains Statement I.**
Final Answer : option (1)

59)

Explain Question : Match the given columns.
Concept : This question is based on qualitative analysis of cations.
Solution :
Pb^{2+} (aq) + $2Cl^-$ (aq) → $PbCl_2$ (s)
Pb^{2+} (aq) + H_2S(g) → PbS(s) + $2H^+$ (aq) Black precipitate of PbS forms.
Fe^{3+}(aq) + $3OH^-$ (aq) → $Fe(OH)_3$(s)
A reddish-brown precipitate forms.
When Ni^{2+} (nickel(II) ions) react with dimethylglyoxime (DMG) solution, a characteristic bright red precipitate of nickel dimethylglyoxime complex forms.
Zn^{2+}(aq) + $4CN^-$(aq) → $[Zn(CN)_4]^{2-}$
The product is a tetra-cyanidozincate(II) complex : $[Zn(CN)_4]^{2-}$, which is colorless and soluble in water.
Final Answer: option (1)

60)

Pb^{+2} → Group I
Fe^{+3} → Group III
Ni^{+2} → Group IV
Mg^{+2} → Group VI

61)

Use formula :

mole (n) = $\frac{wt}{M_w} = \frac{\text{No of particle}}{N_A}$

$$= \frac{\text{Volume}}{22.4}$$

62)

$MnO_4^- \rightarrow Mn^{2+}$
V.F. ⇒ 5
Means 5 mole electron required for reduction of 1 mol MnO_4^-
for 2 mole MnO_4^- = 10 mole electron required
means 10 F charge

63) Based on fact

64) $\Lambda_m^0(HA) = \Lambda_m^0(HCl) + \Lambda_m^0(NaA) - \Lambda_m^0(NaCl)$
= 425.9 + 100.5 - 126.4
= 400 S cm^2 mol^{-1}

$$\Lambda_m = \frac{1000K}{M} = \frac{1000 \times 5 \times 10^{-5}}{10^{-3}} = 50 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{50}{400} = 0.125$$

65) We know, Arrhenius equation

$$\ln K = \ln A - \frac{E_a}{RT}$$

y = C + mx

$$\text{So, slope} = -\frac{E_a}{R} = -1 \times 10^4 \text{ K}$$

So, $E_a = 10^4$ R
$= 10^4 \times 8.3$
$= 8.3 \times 10^4$ J mol^{-1}

66) $K_{sp(Cr(OH)_3)} = 27s^4$

67)

$i = 1 - \alpha + \alpha n$

68)

Based on facts

69) $\Delta T_b = 101.08^\circ - 100^\circ = 1.08^\circ C$
$\Delta FP = 1.80°C$

$$\frac{K_b}{K_f} = 0.3$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b}{K_f} \times \frac{mi_f}{mi_b}$$
$i_f = 1$

70)

HCl is S.A.
NaCl is neutral
CH_3COONa is salt of WASB
NaOH is SB

71)

In free expansion
w = 0
$\Delta S > 0$

72)

$\Delta H_{solution} = \Delta H_{H.E.} + \Delta H_{L.E.}$

73)

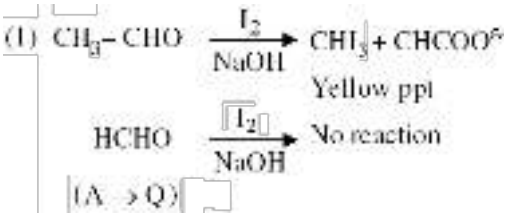
E → +ve (spontaneous)

74) $E = \frac{hc}{\lambda} = h\nu = hc\bar{\nu}$

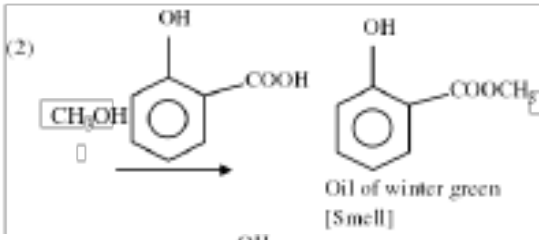
75) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$
2 mol MnO_4^- required ⇒ 16 mol H^+
and 1 mol MnO_4^- required ⇒ 8 mol H^+

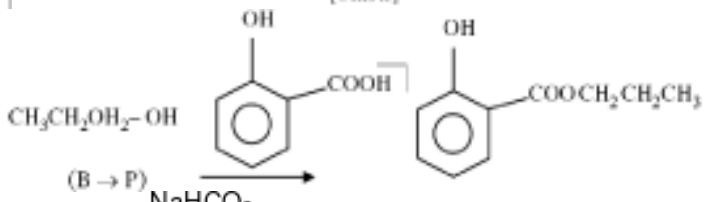
76)

Explaining the Question: Question is asking correct match of compound given in column I with distinguishing reagent column II.
Concept: This question is based on Practical organic chemistry
Solution:



(2)





(3) $PhOH \xrightarrow{NaHCO_3}$ No reaction

$CH_3COOH \xrightarrow{NaHCO_3} CH_3CO\overset{\ominus}{O}\overset{\oplus}{N}a + CO_2 + H_2O$

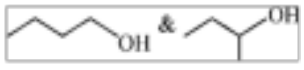
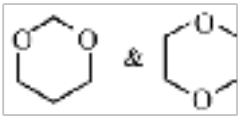
(C → S)

(4) Ketone does not react with tollen's reagent but aldehyde give silver mirror ppt with $AgNO_3$ + NH_4OH (Tollen's reagent) (D → R)

Final Answer : 3

77)

Explain the Question : Determining which pair of compounds are not position isomers.
Concept : This question is based on position isomerism.
Solution :

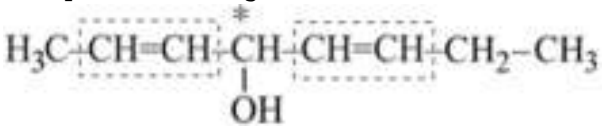
Given compounds	Remark
1  Butan-1-ol Butan-2-ol	Position isomers.
2  1,3-dioxane 1,4-dioxane	Position isomers.
3 $CH_3O-CH_2-CH_2-CH_3$ & $CH_3-O-\underset{CH_3}{CH}-CH_3$ 1-methoxypropane 2-methoxypropane	Position isomers or metamer.
4 	Both are chain isomers.

Final answer : 4

78)

Explain Question: How many stereoisomers are possible for the given compound.
Concept: Number of stereoisomers.

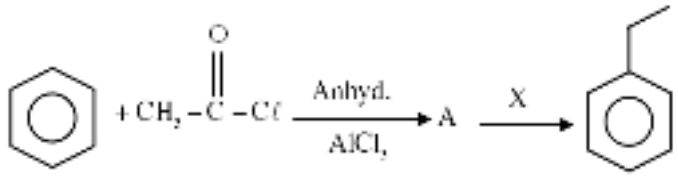
Solution:
Compounds having dissimilar ends (n=3)



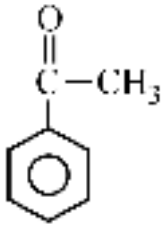
Total no. of stereoisomers = $2^n = 2^3 = 8$
Final answer: 2

79)

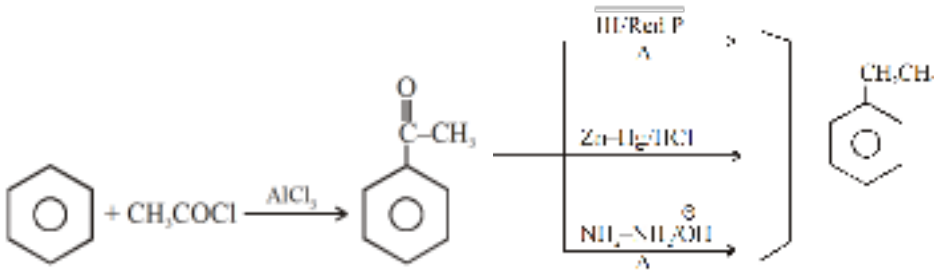
Explaining the Question :-



reagent 'X' can be :



Concept:- This question is based on Acylation of benzene, reduction of .
Solution :-

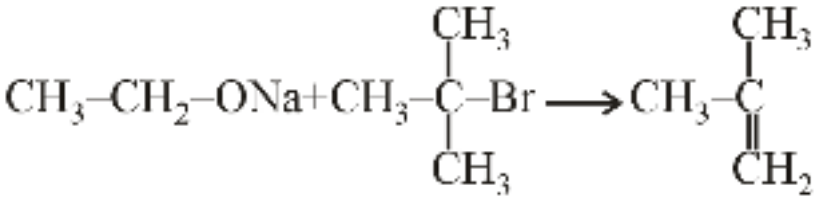


Final Answer :- 4

80) **Explain Question:** Identify which reaction does not produce an ether as the product.

Concept: Williamson Ether Synthesis, S_N2 Mechanism, Steric Hindrance

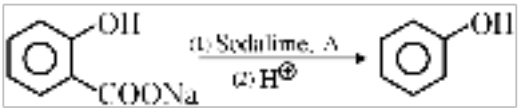
Solution:



3° alkyl halide gives elimination reaction, not substitution.

Final answer: Option (3)

81) $Ph-Cl \xrightarrow[H^{\oplus}]{NaOH\ 623K,\ 300atm,} Ph-OH$



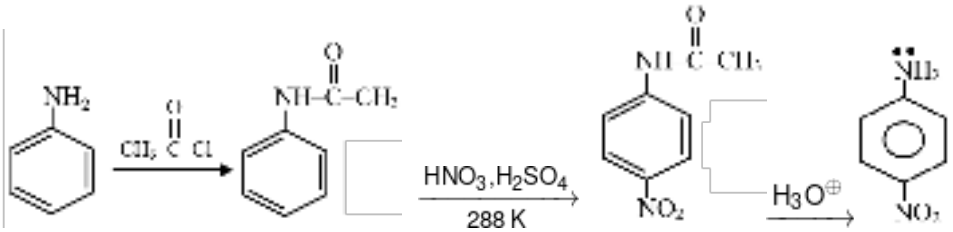
$C_6H_5MgBr \xrightarrow[H^{\oplus}]{H_2O} C_6H_6$

$C_6H_5\overset{\oplus}{N}_2\overset{\ominus}{Cl} \xrightarrow[Boil]{H_2O} C_6H_5-OH$

82) $CH_3-CH_2-\overset{\oplus}{N}\equiv\overset{\ominus}{C} \xrightarrow{H_2/Pd-C} CH_3-CH_2-NH-CH_3$

(2° - Amine)

83) **Question Explanation :**
In the given chemical reactions, the major product is :
Concept :
Acetylation followed by nitration then hydrolysis.
Solution :



Final Answer :
Option (2)

84)

Question Explanation : Asking about the compound which can given Haloform test

Concept : Haloform test

Solution :
Compounds that give a positive iodoform test contain either:

A. Methyl ketone group $CH_3C(=O)-$

B. Ethanol or secondary alcohol with the structure $CH_3CH(OH)-$ (which can be oxidized to a methyl ketone during the reaction).

Therefore, $CH_3CH(OH)CH_2CH_3$ gives the haloform test

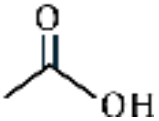
Final Answer : Option (2)

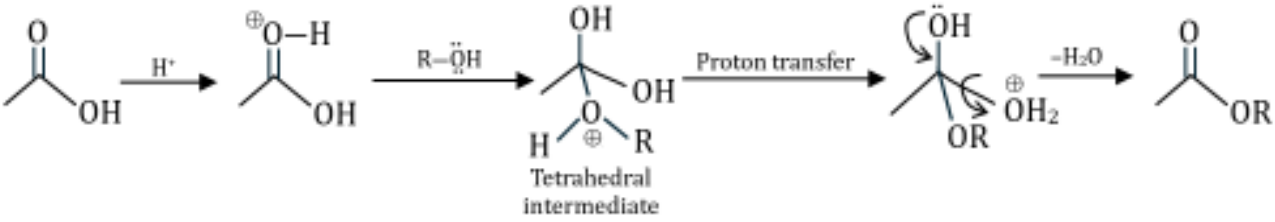
85)

Question Explanation : Assertion and reason

Concept : Esterification

Solution :

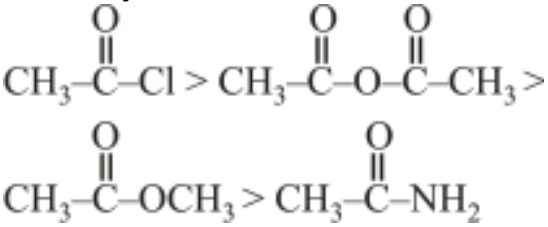
 is less reactive towards NAR, but protonation of carbonyl activates it towards NAR



Final Answer : Option (1)

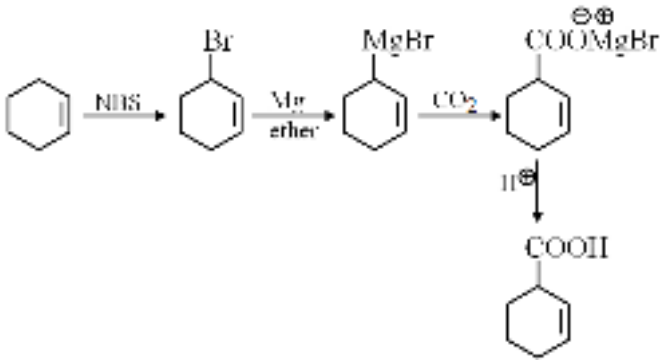
86)

Reactivity order :



due to leaving group ability.

87)



88)

Fact

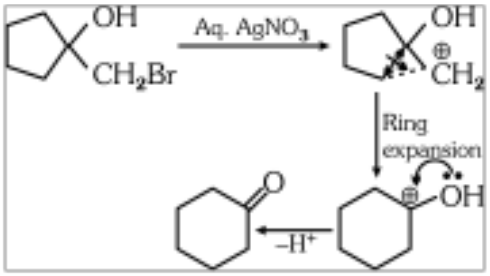
89) $0.1g \rightarrow NH_3 \rightarrow 30ml\ \frac{N}{5}H_2SO_4 \rightarrow 20ml\frac{N}{10}NaOH$

$20ml \times \frac{N}{10} = 2ml\ N\ NaOH = 2ml\ N\ H_2SO_4$

Remaining acid = 2ml N H_2SO_4

$\Rightarrow 30ml \times \frac{N}{5} - 2ml\,N \Rightarrow 4ml\,N\,H_2SO_4$

$\%N = \frac{1.4}{W} \times NV \Rightarrow \frac{1.4}{0.1} \times 4 = 56\%$

90) 

BIOLOGY

91) NCERT Page No. 7

92)

NCERT Pg. # 15

93) NCERT Page No. 15-16

94) NCERT-XI, Pg.# 19-21

95) NCERT Pg. # 17

96) NCERT Pg. # 29, 24, 81

97) NCERT Page No. 32

98) NCERT Pg. # 33

99)

NCERT (old) Pg. # 79, 80, 81

Name of genera	No. of Carpel(s) in their gynoecium.
Pisum	1
Solanum	2

Allium	3
Hibiscus	5

100) NCERT Pg. # 60

101) NCERT XI Pg.No.#70-71(E)

102) NCERT XII Pg # 76

103) NCERT Page No. 72-75

104) NCERT Pg. # 22, 23

105) NCERT Pg. No. 9

106) NCERT-XII Pg. # 13, 14, 29, 33

107) NCERT Pg # 117

108) NCERT, Pg. # 109,112

109) NCERT-XI, Pg # 109

110)

NCERT-XI, Pg. # 137, 138, 147, 150

111) NCERT Pg # 143(E) 144, 145 (H)

112) NCERT pg no # 158, 163

113) NCERT, Pg. # 156-160

114) NCERT Pg. # 248

115) NCERT Pg. # 172

116) NCERT Pg. # 175-177

117) NCERT, Pg.# 71

118) NCERT Pg. # 74, 75, 76

119) NCERT Pg. No. # 62

120) NCERT Pg. No # 80

121) NCERT Pg. # 67

122) NCERT, Pg.# 104

123) NCERT Pg # 80–83

124)

NCERT Pg. # 92-95

125) NCERT, Pg # 95

126) NCERT Page No. 90

127) NCERT Pg. # 96

128) NCERT, Pg. # 201

129) NCERT, Pg. # 194

130) NCERT XII Pg.# 208

131) NCERT-XII, Pg # 209-210

132) NCERT Pg. # 212, 213, 214

133)

NCERT Pg. # 224, 225

134) NCERT, Pg # 218

135) NCERT, Pg. # 223

136)

NCERT (Old), Pg. # 113

137)

NCERT, Pg. # 39

138)

NCERT, Pg. # 46

139)

NCERT, Pg. # 46

140) NCERT-XI, Pg. No. # 49

141)

NCERT, Pg. # 48

142)

Module 4, Pg. # 94

143) NCERT-XI Pg. # 116

144) NCERT Pg # 193-215

145) NCERT (XI) Page No. 101

146)

NCERT (XI) Page No. 96, 100

147) NCERT (XI) Page No. 95-96

148) NCERT-XI, page No. - 126

149) NCERT-XI, page No. - 123-125

150)

NCERT(Old), Pg. # 104

151) NCERT Pg. # 199, 200

152)

NCERT, Pg. # 187

153) NCERT, Pg. # 187, 189, 190

154) NCERT Pg. # 208

155) NCERT XII, Page # 211

156) NCERT Pg. # 230, 231

157) NCERT Pg. 230,231

158) NCERT, Pg # 243

159) NCERT, Pg. # 243, 246

160) NCERT Page No. #242, 243, 244

161) NCERT, Pg # 245,246

162)

NCERT, Pg. # 223

163) NCERT XI Pg.#225

164) NCERT Pg. # 38

165) NCERT Pg. # 37

166) NCERT Page no. 35-38

167) NCERT Pg. # 37

168) NCERT XII (E) Page no. 38
NCERT XII (H) Page no. 41, 42

169) NCERT (XI) Page No. 46, 47, 48

170) NCERT, Pg. # 129-143

171) NCERT (XII) Page#135/147(H) Para:7.2.2

172) NCERT, Pg. # 129, 130

173) NCERT Pg. No. 156

174) NCERT Pg. # 118, 119

175) NCERT Pg.#124

176) NCERT-XII, Pg # 174

177)

NCERT-XII, Pg. # 168

178) NCERT - 208 (12.1)

179) Page No. – 211

180) NCERT XII, Pg. # 156